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EGSD/ASU

Airport Local Air Quality Studies with OpenALAQS

Supporting European Aviation

EUROCONTROL

open-alaqs@eurocontrol.int

CONSORTIUM COORDINATOR sesar DEPLOYMENT MANAGER

FOUNDING MEMBER sesar JOINT UNDERTAKING

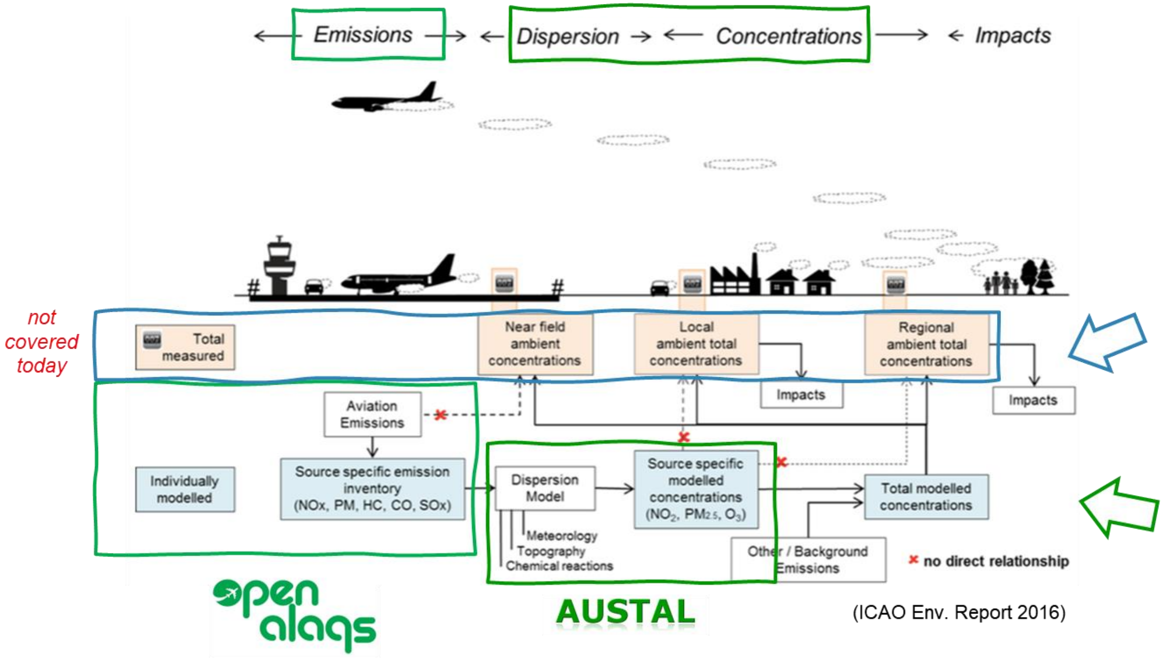
NETWORK MANAGER

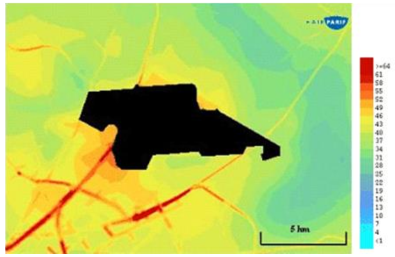
Agenda

- **Introduction**
  - *Key Objectives of the Training Course*
  - *General information about OpenALAQS/QGIS*
- **OpenALAQS:**
  - *Create a New Study*
  - *Define Emission Sources*
  - *Generate Emissions Inventory*
  - *Calculate & visualize emissions*
  - *Useful information about the emissions calculation methods*
- **Dispersion Modeling with AUSTAL**
  - *Data Preparation*
  - *Calculate & visualize concentrations*
- **Q&A**

Key Objectives of the Training Course

- Describe the core features of OpenALAQS for assessing airport emissions
- Explore the modelling capabilities of OpenALAQS
- Interpret results and understand their implications for air quality management at airports

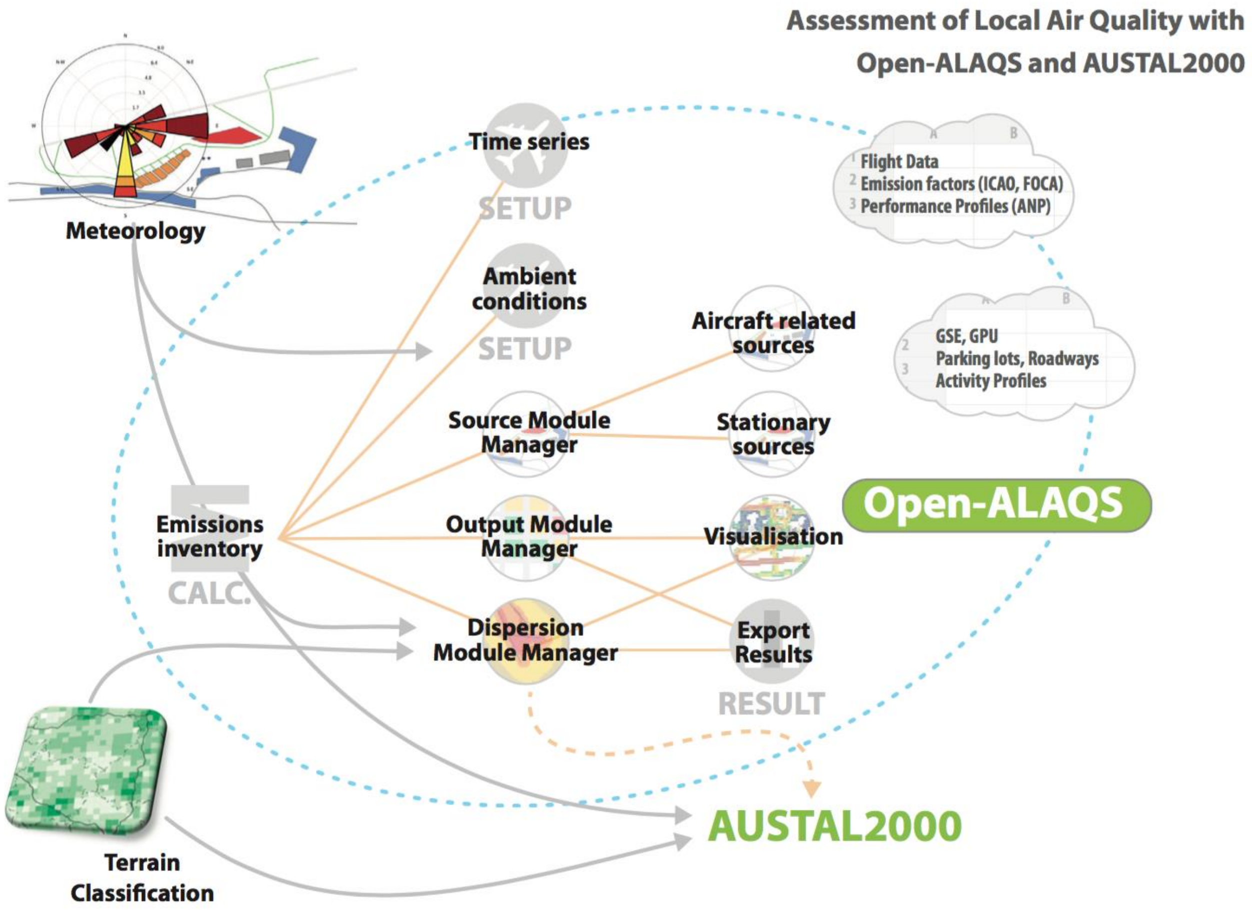




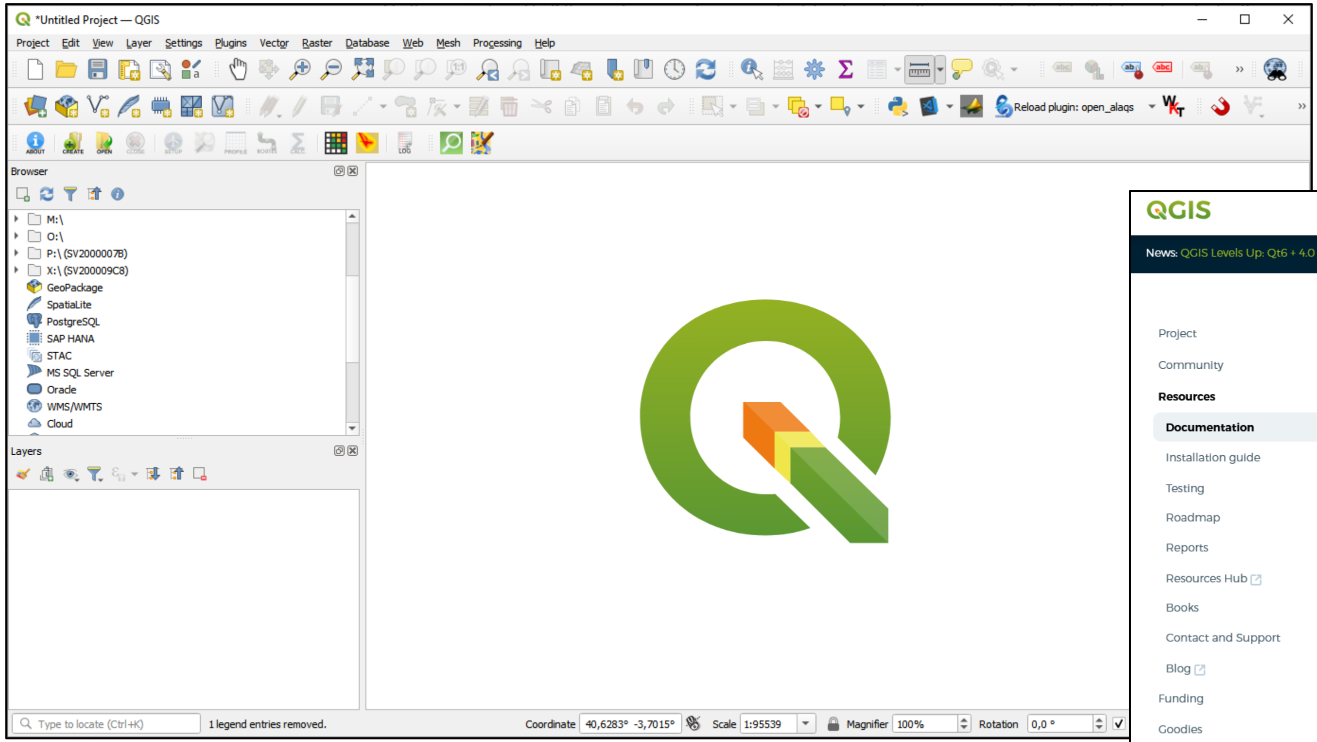
Open-ALAQS

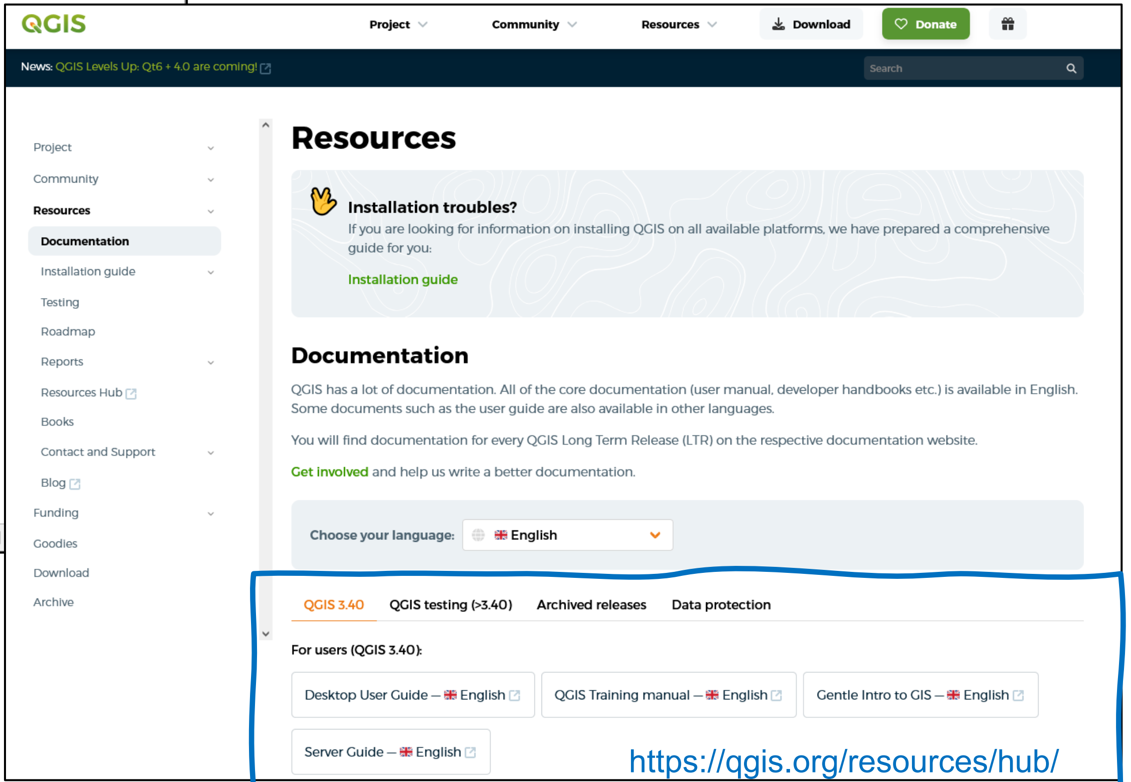
- Developed as plugin for QGIS
- Full LAQ modelling suite (with AUSTAL)
- Modular architecture
- Code publicly available on GitHub

  https://github.com/eurocontrol-asu/open_alaqs



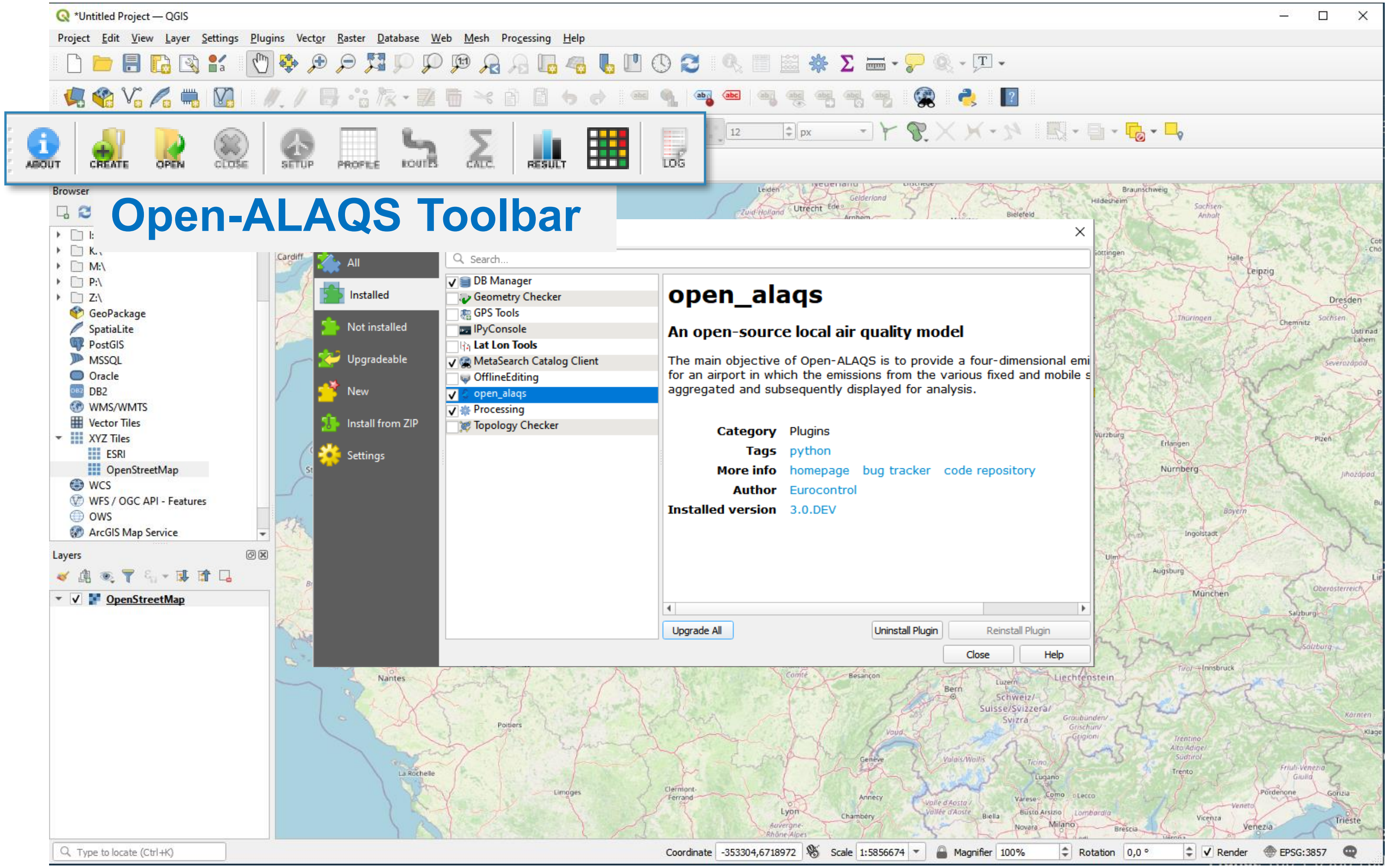
Short introduction to QGIS



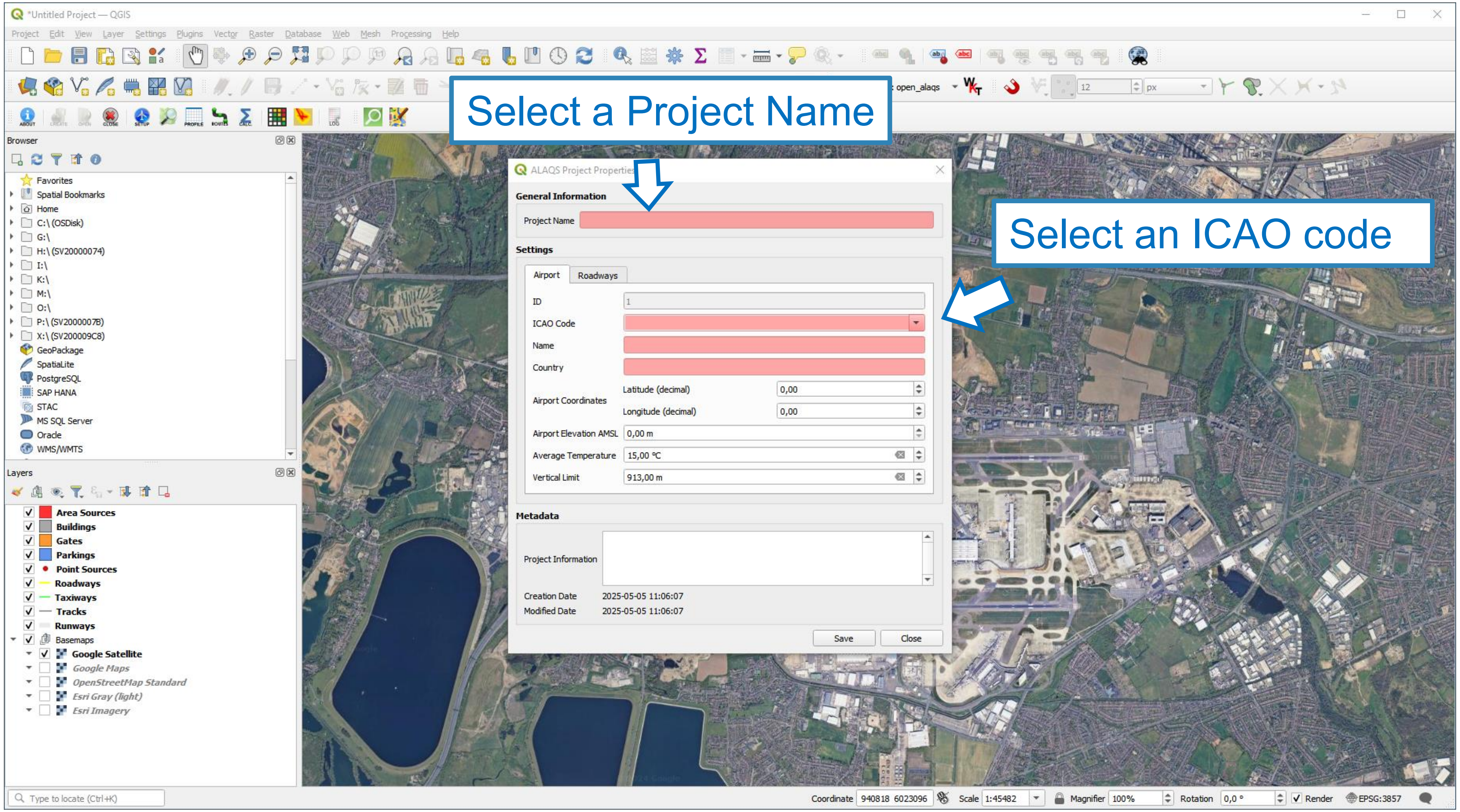


https://qgis.org/resources/hub/

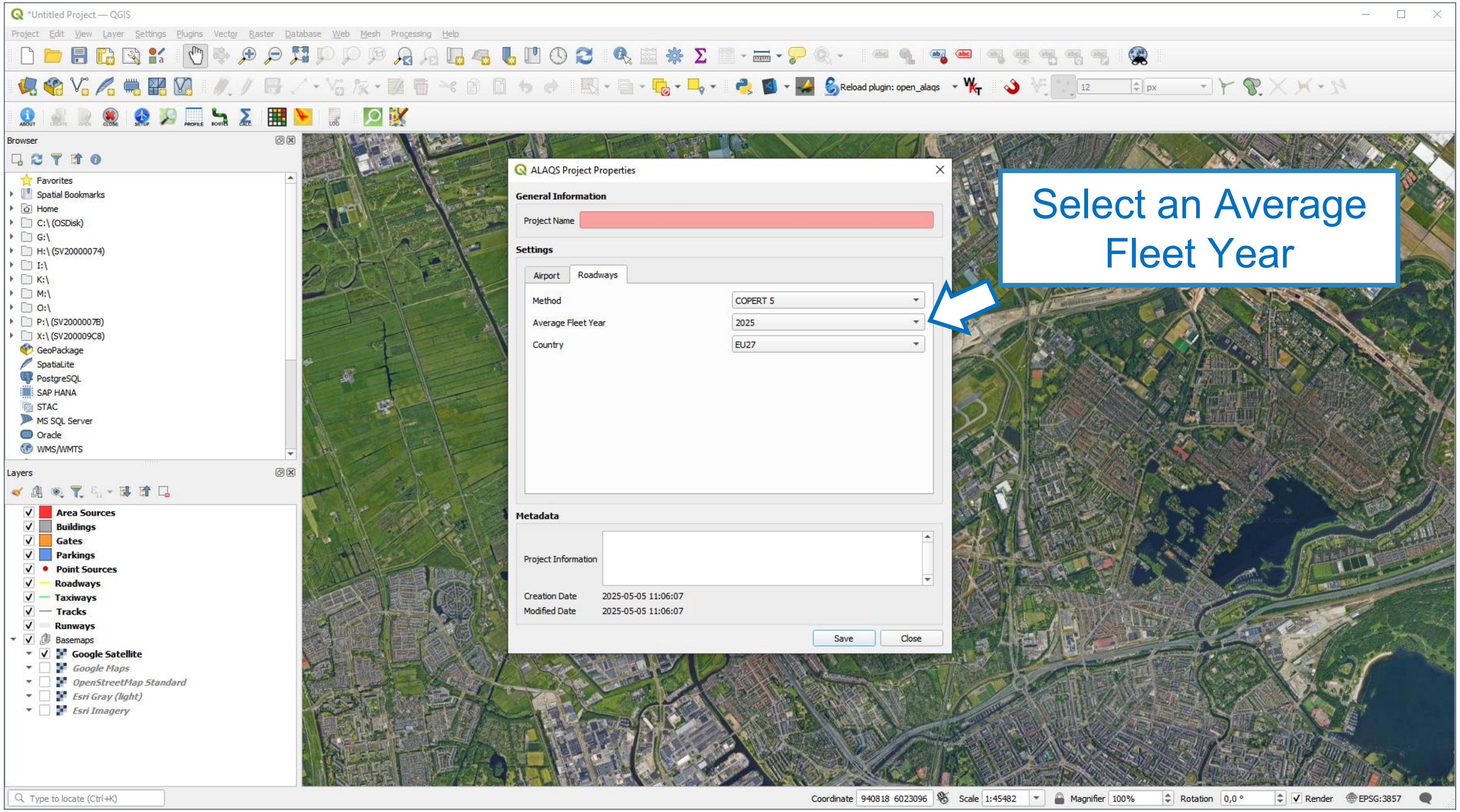
Open-ALAQS / QGIS



Create a new study

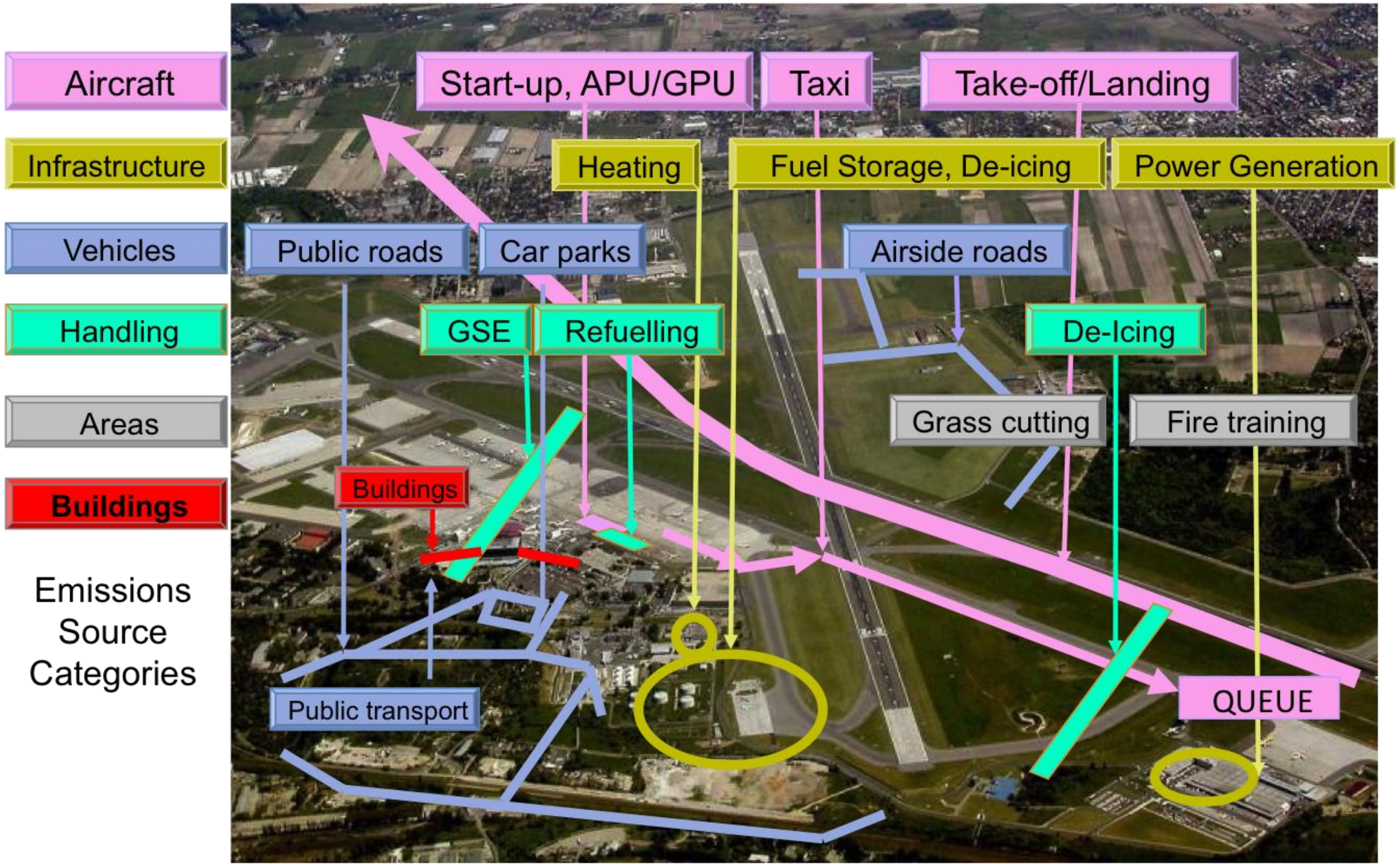


Create a new study



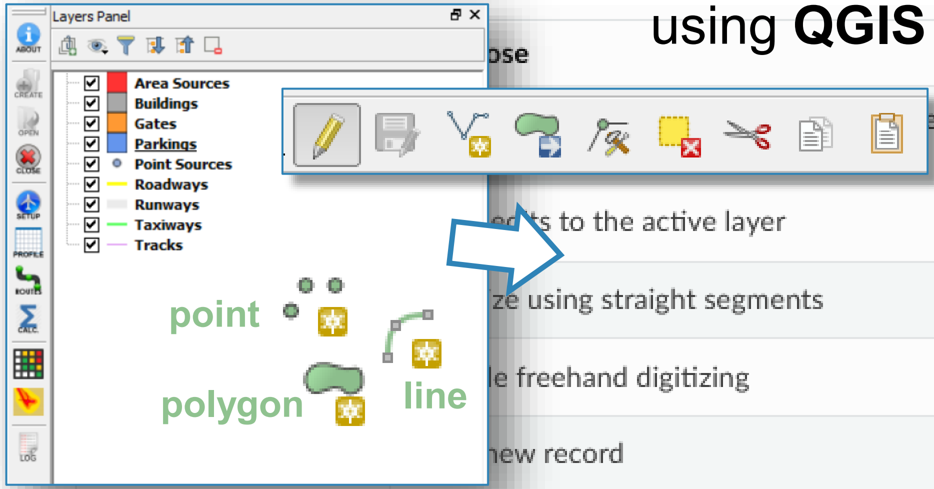
Select an Average Fleet Year

Definition of Emission Sources



Definition of Emission Sources

using **QGIS** native toolbar



Tool	...ose	Tool	Purpose
	...es in all or selected layers		Turn on or off edit status of selected layer(s) based on the active layer status
	...ts to the active layer		
	...ze using straight segments		Digitize using curve lines
	...le freehand digitizing		Digitize polygon of regular shape
	...new record		Add Feature: Capture Point
	Add Feature: Capture Line		Add Feature: Capture Polygon
	Vertex Tool (All Layers)		Vertex Tool (Current Layer)
Show Vertex Editor	Set whether the vertex editor panel should auto-open		Modify the attributes of all selected features simultaneously
	Delete Selected features from the active layer		Cut Features from the active layer
	Copy selected Features from the active layer		Paste Features into the active layer
	Undo changes in the active layer		Redo changes in active layer

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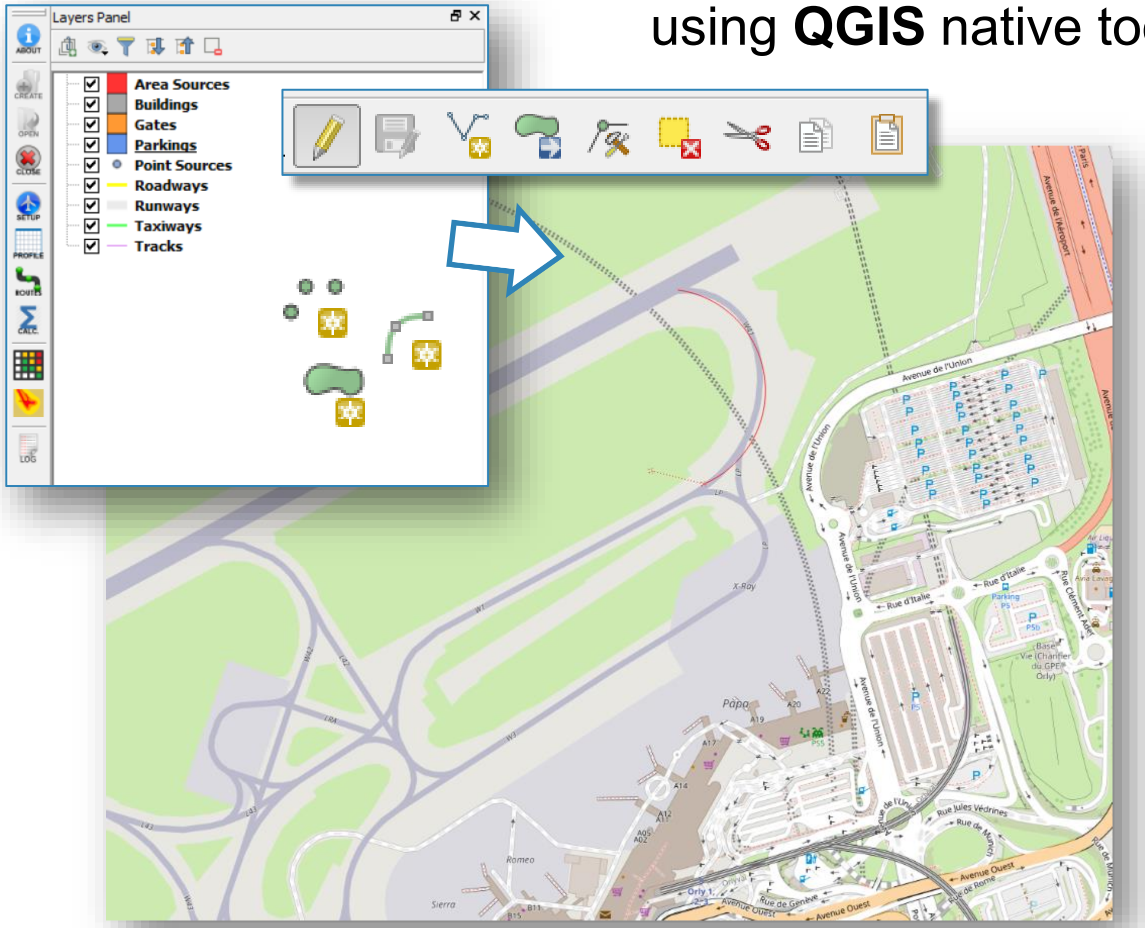
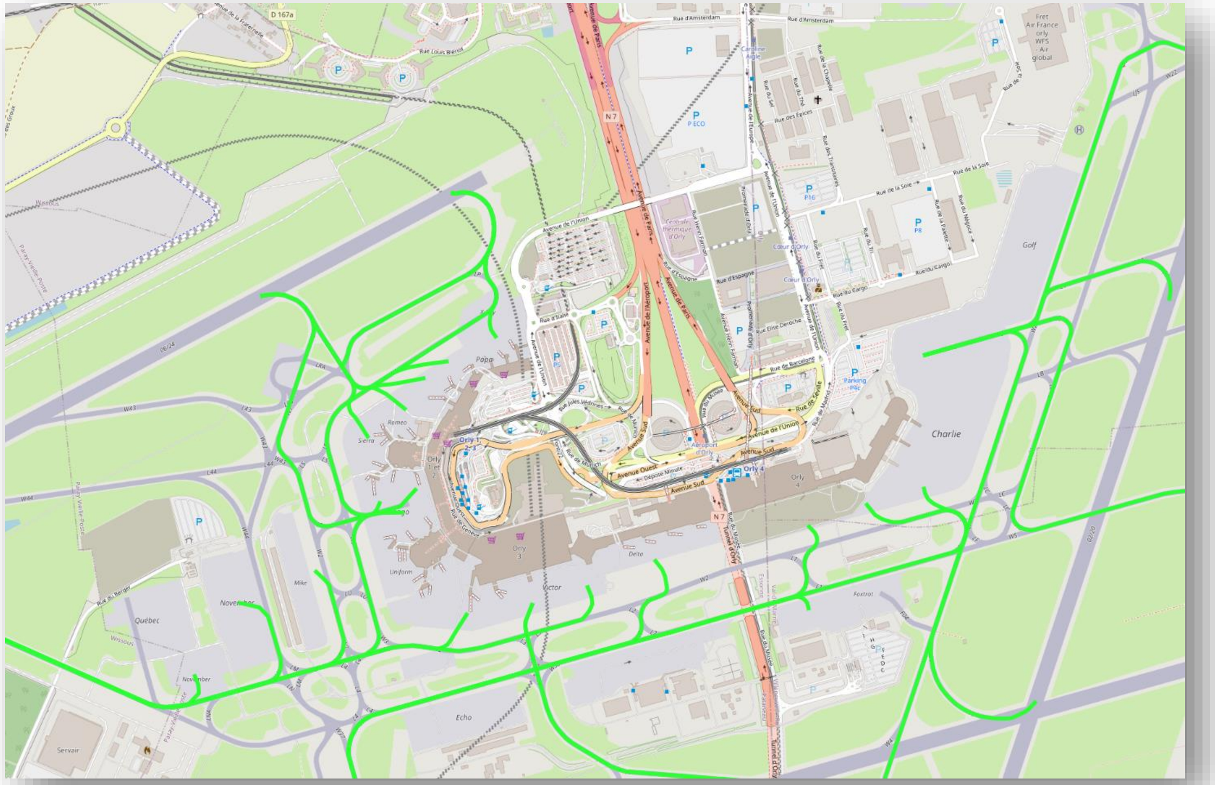
Definition of Emission Sources

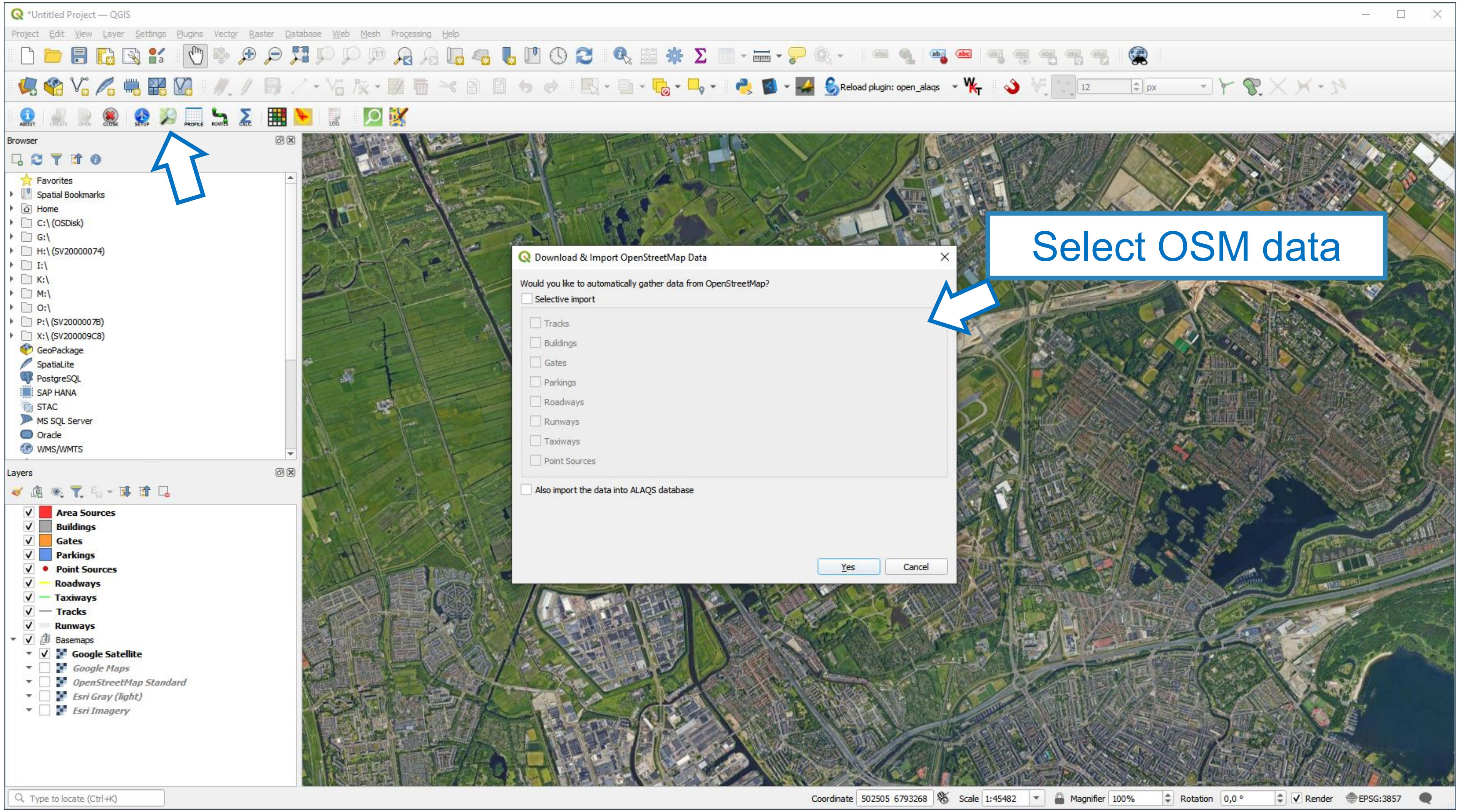
using **QGIS** native toolbar





Import OpenStreetMap data

(Optional)



Import OpenStreetMap data

(Exercise)

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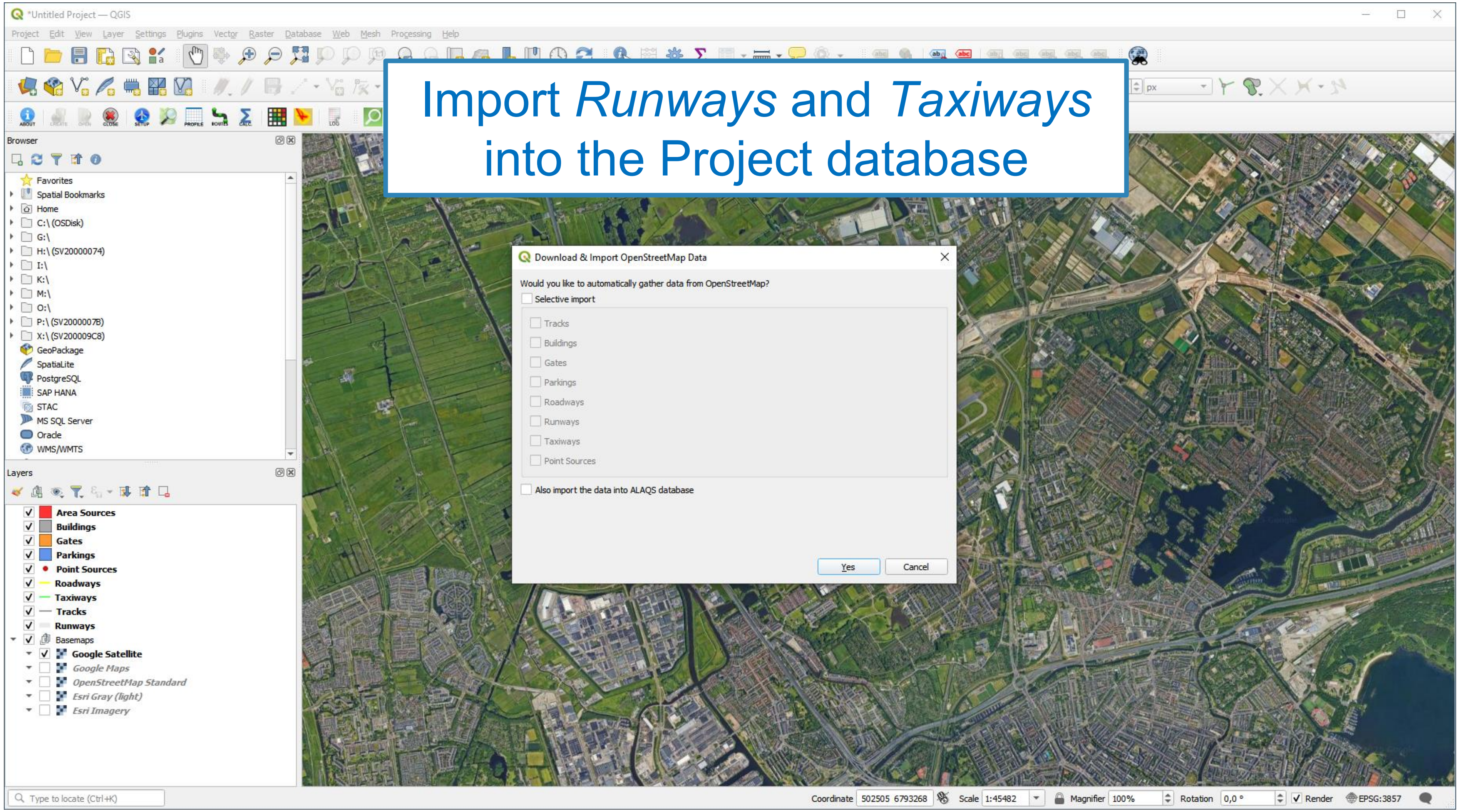
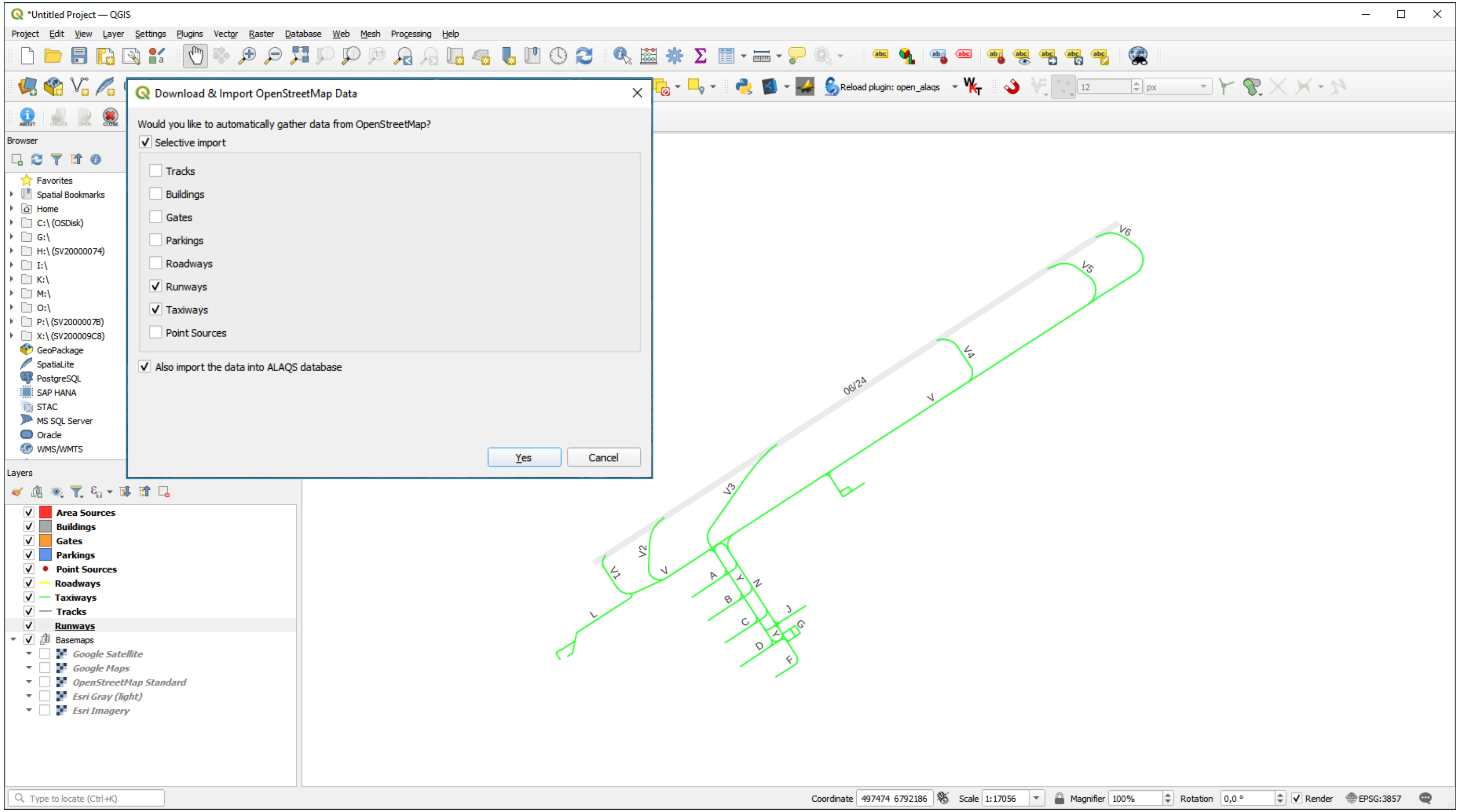
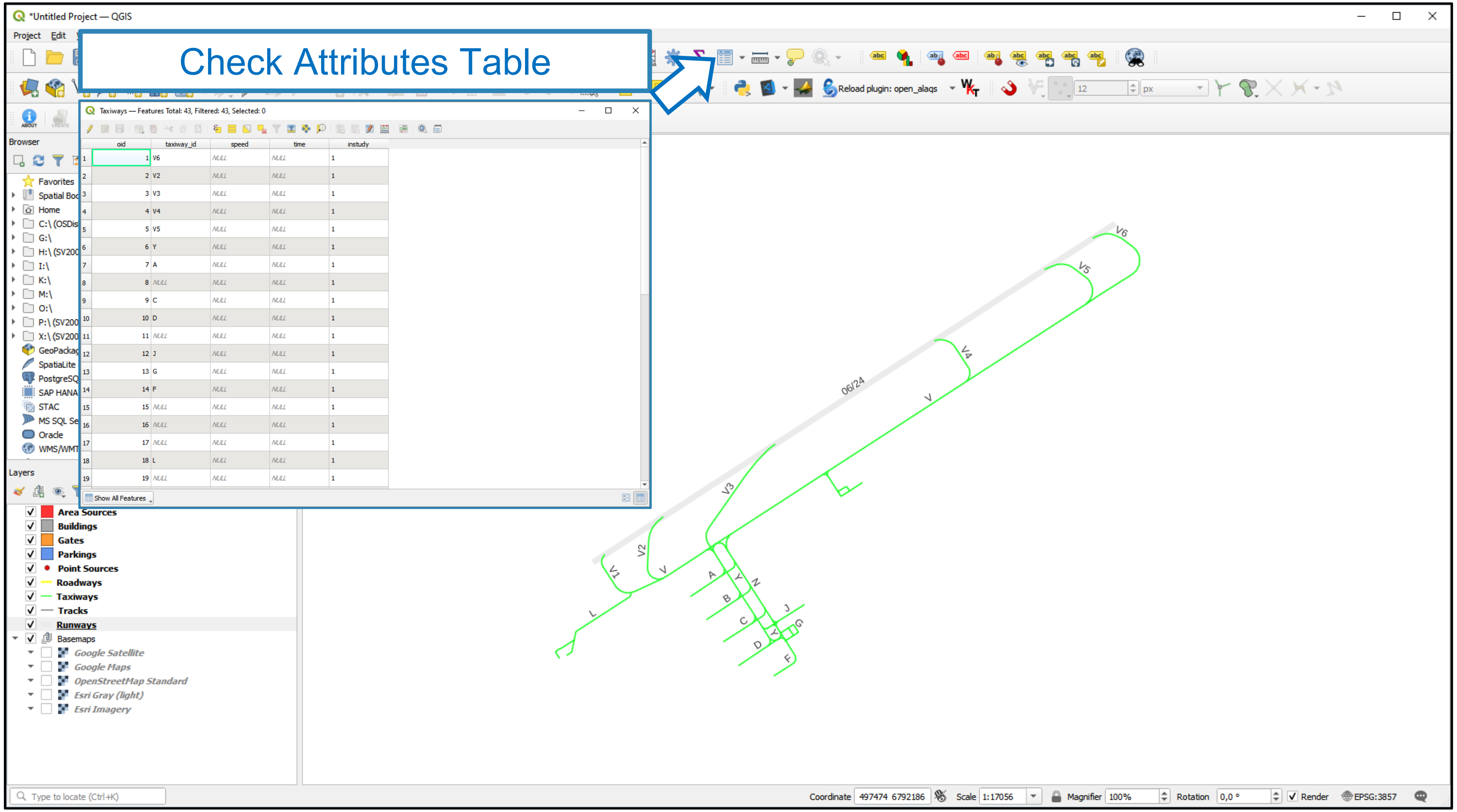


Import OpenStreetMap data

(Exercise)



Import OpenStreetMap data



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Import OpenStreetMap data

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Ensure all imported Features have valid attribute values and names

Runways and Taxiways

Taxiways — Features Total: 43, Filtered: 43, Selected: 0

	oid	taxiway_id	speed	time	instudy
1	1	V6	NULL	NULL	1
2	2	V2	NULL	NULL	1
3	3	V3	NULL	NULL	1
4	4	V4	NULL	NULL	1
5	5	V5	NULL	NULL	1
6	6	Y	NULL	NULL	1
7	7	A	NULL	NULL	1
8	8	NULL	NULL	NULL	1
9	9	C	NULL	NULL	1
10	10	D	NULL	NULL	1
11	11	NULL	NULL	NULL	1
12	12	J	NULL	NULL	1
13	13	G	NULL	NULL	1
14	14	F	NULL	NULL	1
15	15	NULL	NULL	NULL	1
16	16	NULL	NULL	NULL	1
17	17	NULL	NULL	NULL	1
18	18	L	NULL	NULL	1
19	19	NULL	NULL	NULL	1

Show All Features

Import OpenStreetMap data

Ensure all imported Features have valid attribute values and names

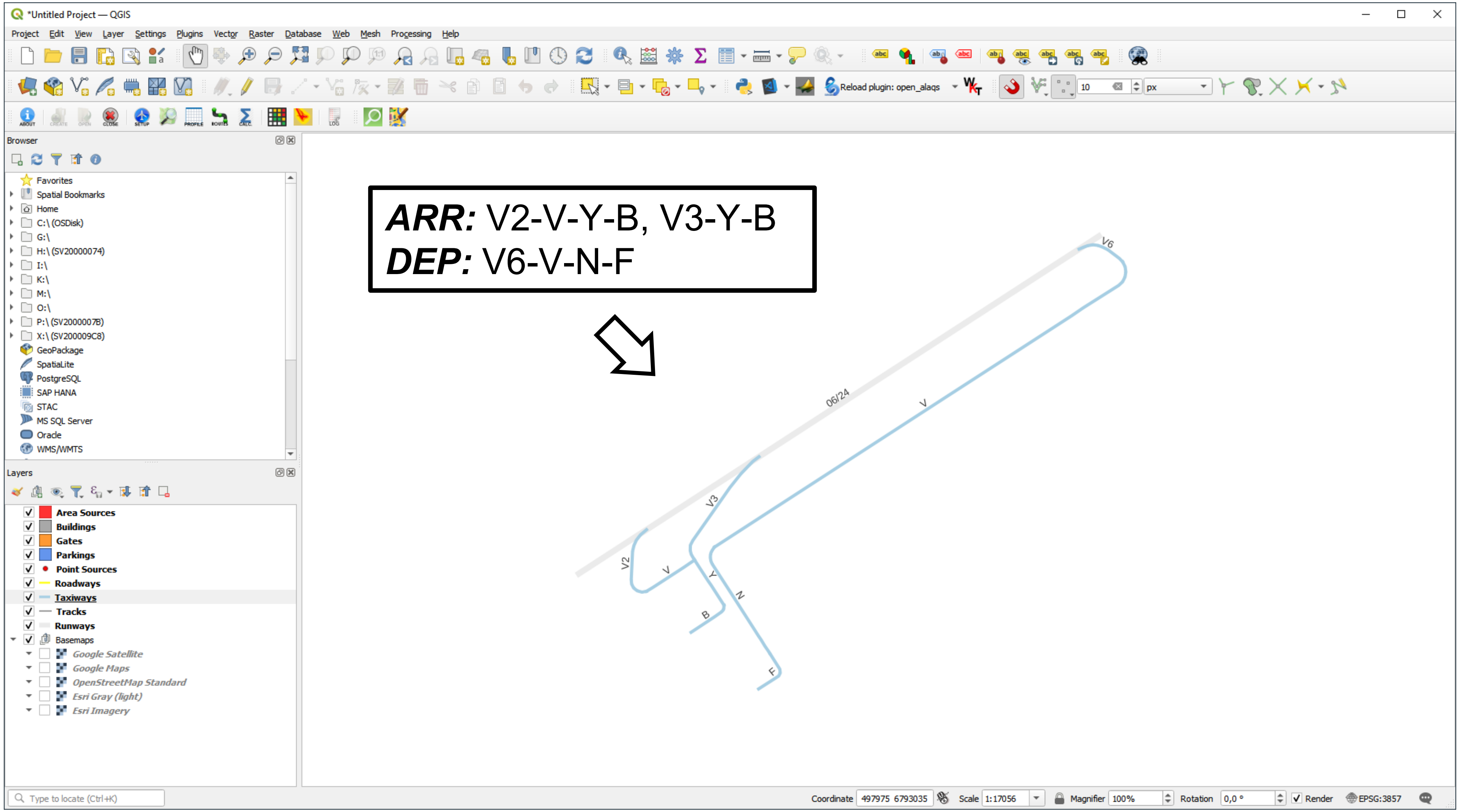
Taxiways — Features Total: 43, Filtered: 43, Selected: 0

	oid	taxiway_id	speed	time	instudy
1	1	V6	NULL	NULL	1
2	2	V2	NULL	NULL	1
3	3	V3	NULL	NULL	1
4	4	V4	NULL	NULL	1
5	5	V5	NULL	NULL	1
6	6	Y	NULL	NULL	1
7	7	A	NULL	NULL	1
8	8	NULL	NULL	NULL	1
9	9	C	NULL	NULL	1
10	10	D	NULL	NULL	1
11	11	NULL	NULL	NULL	1
12	12	J	NULL	NULL	1
13	13	G	NULL	NULL	1
14	14	F	NULL	NULL	1
15	15	NULL	NULL	NULL	1
16	16	NULL	NULL	NULL	1
17	17	NULL	NULL	NULL	1
18	18	L	NULL	NULL	1
19	19	NULL	NULL	NULL	1

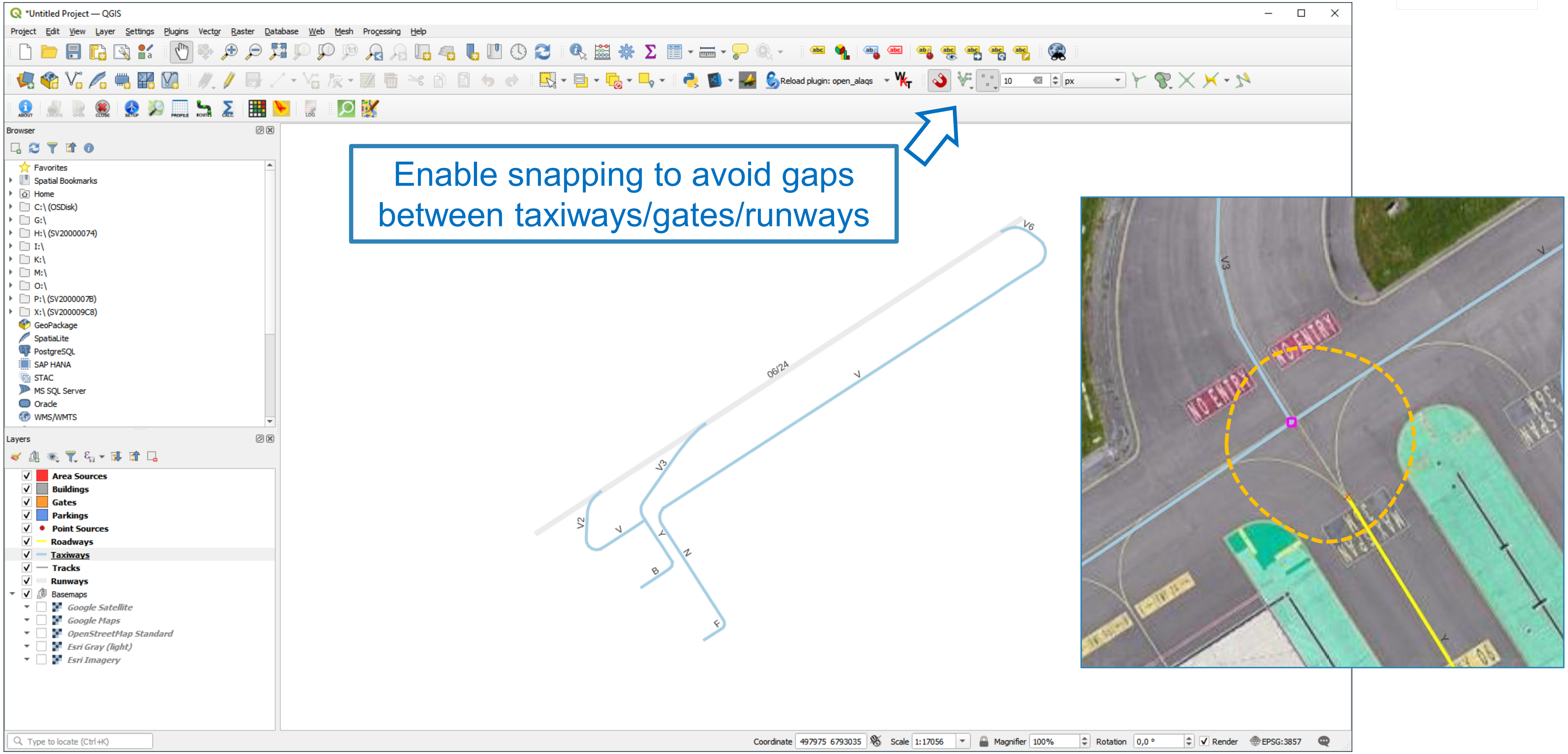
Show All Features

- Ensure that there are no duplicate IDs
- Typical speed: *30 km/h*
- Time: calculated from speed and taxiway segment length

Simplify taxiways



Simplify taxiways



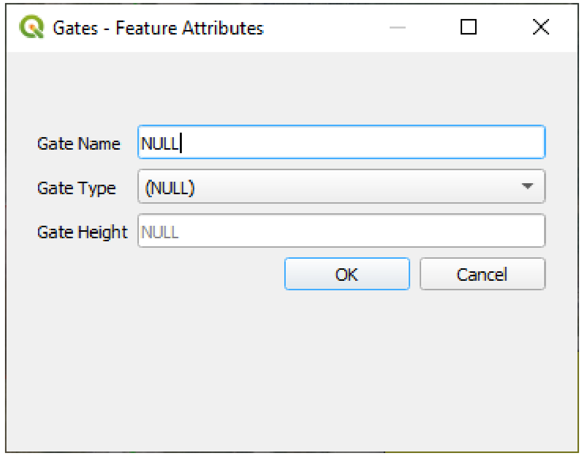
Enable snapping to avoid gaps between taxiways/gates/runways

Add Gates

An airport gate refers to a designated location at an airport where aircraft park for boarding and disembarking passengers, loading/unloading cargo, and receiving services like refuelling, catering, and maintenance.

In OpenALAQS, gates are represented as polygons. Each gate can encompass several aircraft stands. The more stands grouped together within a single gate area, the less data preparation is needed (e.g., fewer taxi routes to define). However, if the gate area is too large, it might no longer accurately represent the location of the emissions.

Calculating gate emissions requires establishing the sum of four emission sources: GSE (Ground Support Equipment), GPU (Ground Power Unit), APU (Auxiliary Power Unit) and MES (Main Engine Start).



When adding a gate, the following information is required:

- Gate type (PIER, REMOTE or CARGO)
- Gate height `not yet fully implemented`

In OpenALAQS, GSE and GPU emissions factors, expressed in terms of grams of pollutant per hour, is assigned to each gate as a function of:

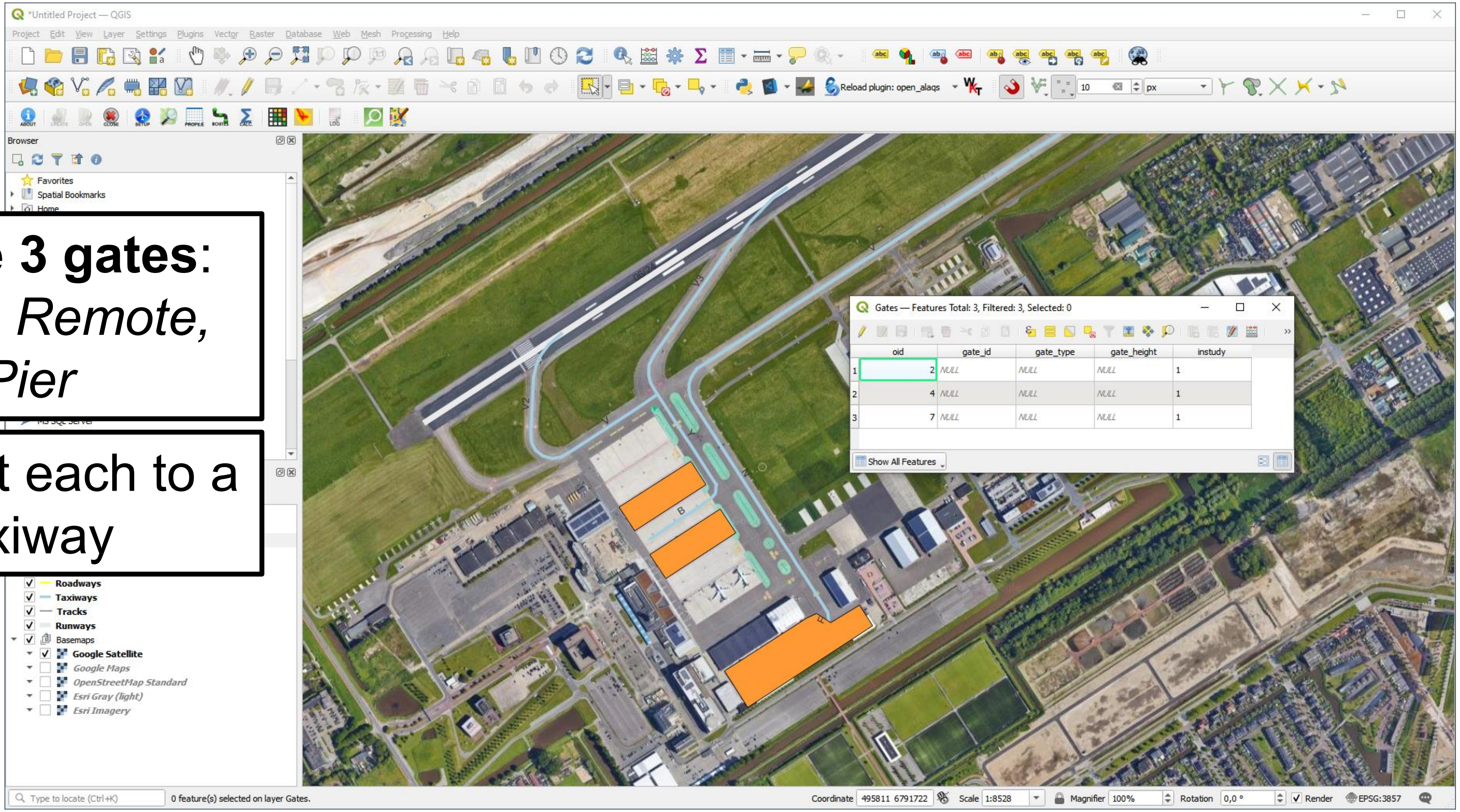
- The gate type (PIER, REMOTE or CARGO)
- The aircraft category (JET BUSINESS/REGIONAL/SMALL/MEDIUM/LARGE,TURBOPROPS,PISTON)
- The operation type (Arrival or Departure)

More information on the corresponding GSE/GPU, APU and MES emission factors and activity time is available in `OpenALAQS database`.

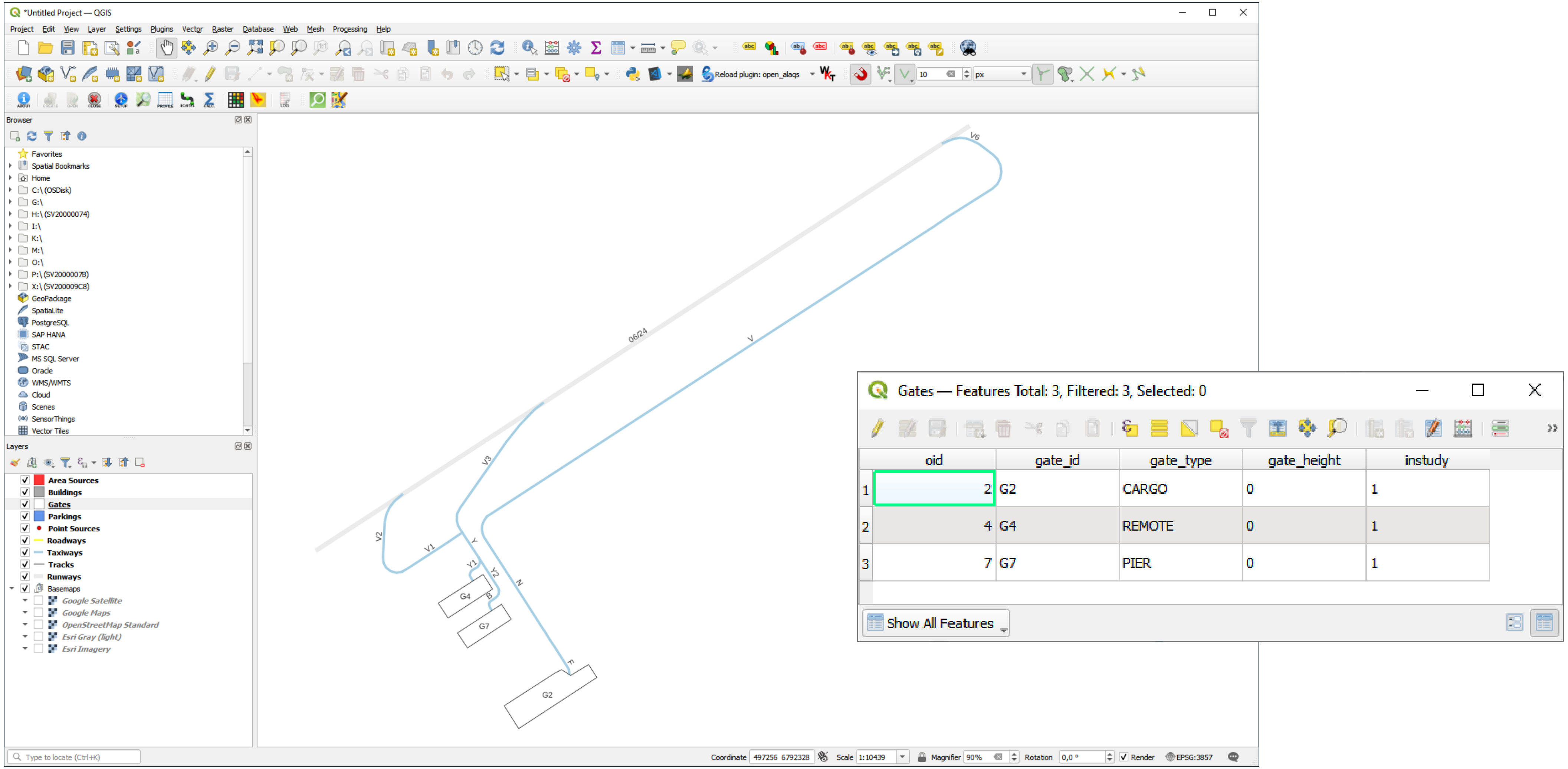
Add Gates

Create 3 gates:
Cargo, Remote, Pier

Connect each to a taxiway



Add Gates

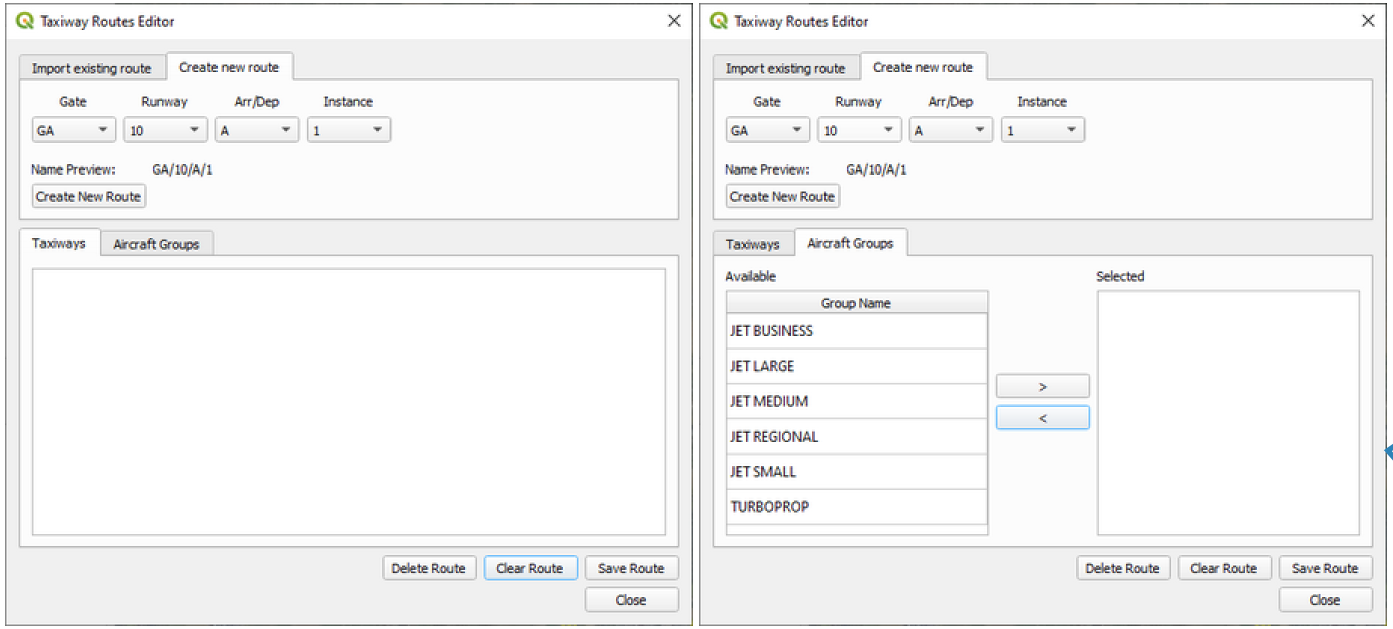


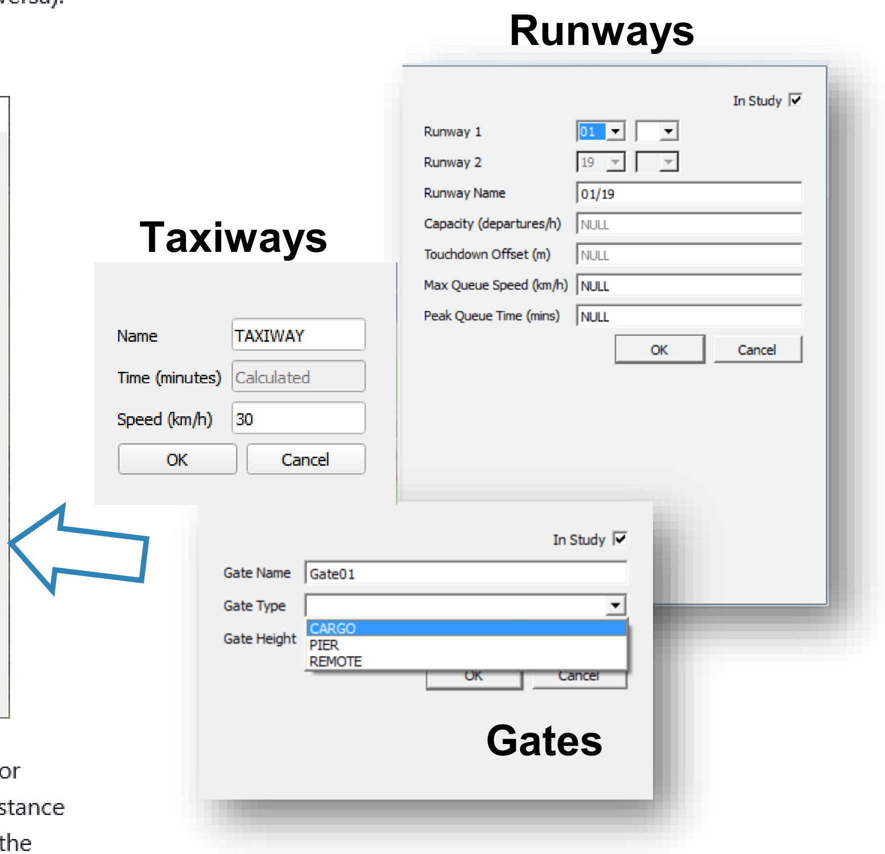
*Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Reload plugin: open_alaqs
Browser
Favorites
Spatial Bookmarks
Home
C:\ (OSDisk)
G:\
H:\ (SV20000074)
I:\
K:\
M:\
O:\
P:\ (SV2000007B)
X:\ (SV200009C8)
GeoPackage
SpatiaLite
PostgreSQL
SAP HANA
STAC
MS SQL Server
Oracle
WMS/WMTS
Cloud
Scenes
SensorThings
Vector Tiles
Layers
Area Sources
Buildings
Gates
Parkings
Point Sources
Roadways
Taxiways
Tracks
Runways
Basemaps
Google Satellite
Google Maps
OpenStreetMap Standard
Esri Gray (light)
Esri Imagery
V6
06/24
V
V3
V2
V1
Y
Y1
Y2
N
G4
G7
F
G2
Gates — Features Total: 3, Filtered: 3, Selected: 0
oid | gate_id | gate_type | gate_height | instudy
1 | 2 | G2 | CARGO | 0 | 1
2 | 4 | G4 | REMOTE | 0 | 1
3 | 7 | G7 | PIER | 0 | 1
Show All Features
Type to locate (Ctrl+K)
Coordinate 497256 6792328
Scale 1:10439
Magnifier 90%
Rotation 0,0 °
Render
EPSG:3857

Define Taxiway Routes

As explained in the section `Taxiways`, taxi-routes describe the operational path of an aircraft between the runway and the gate (or vice versa).

Taxi-routes can be defined using the Taxiway Routes Editor.





To define a taxi route in OpenALAQS, the user has to first create the taxi-route by selecting the gate, runway and operation type (arrival or departure). More than one taxi-routes can be defined for the same combination of gate, runway and operation type, using a different instance number. Once defined, the corresponding taxiway segments have to be selected together with the aircraft groups that can make use of the specific taxi-route.

Define Taxiway Routes

Create 3 taxiroutes: 2 *ARR*, 1 *DEP*

1

Create New Route

Select Gate, Runway, Operation Type

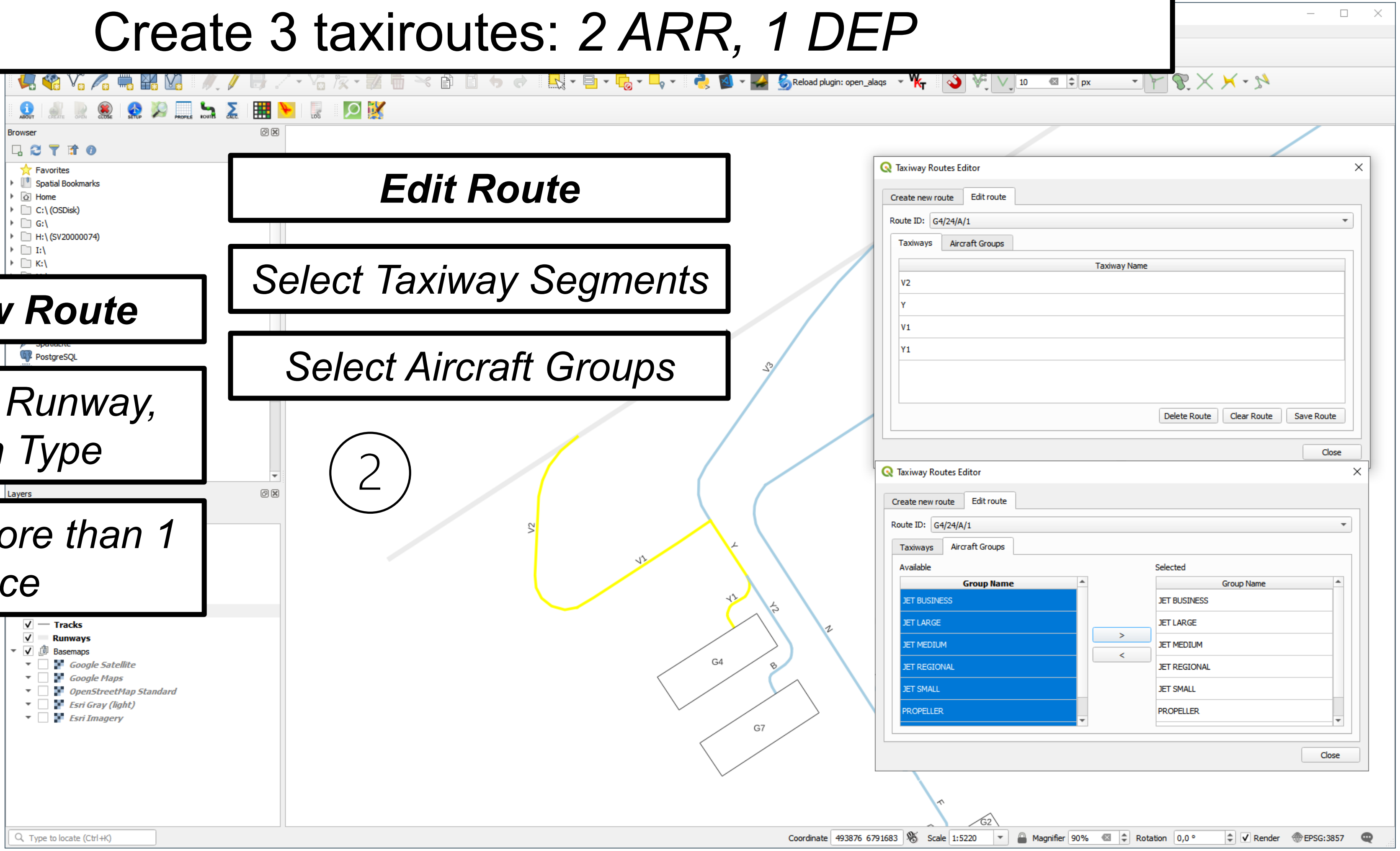
Can create more than 1 instance

2

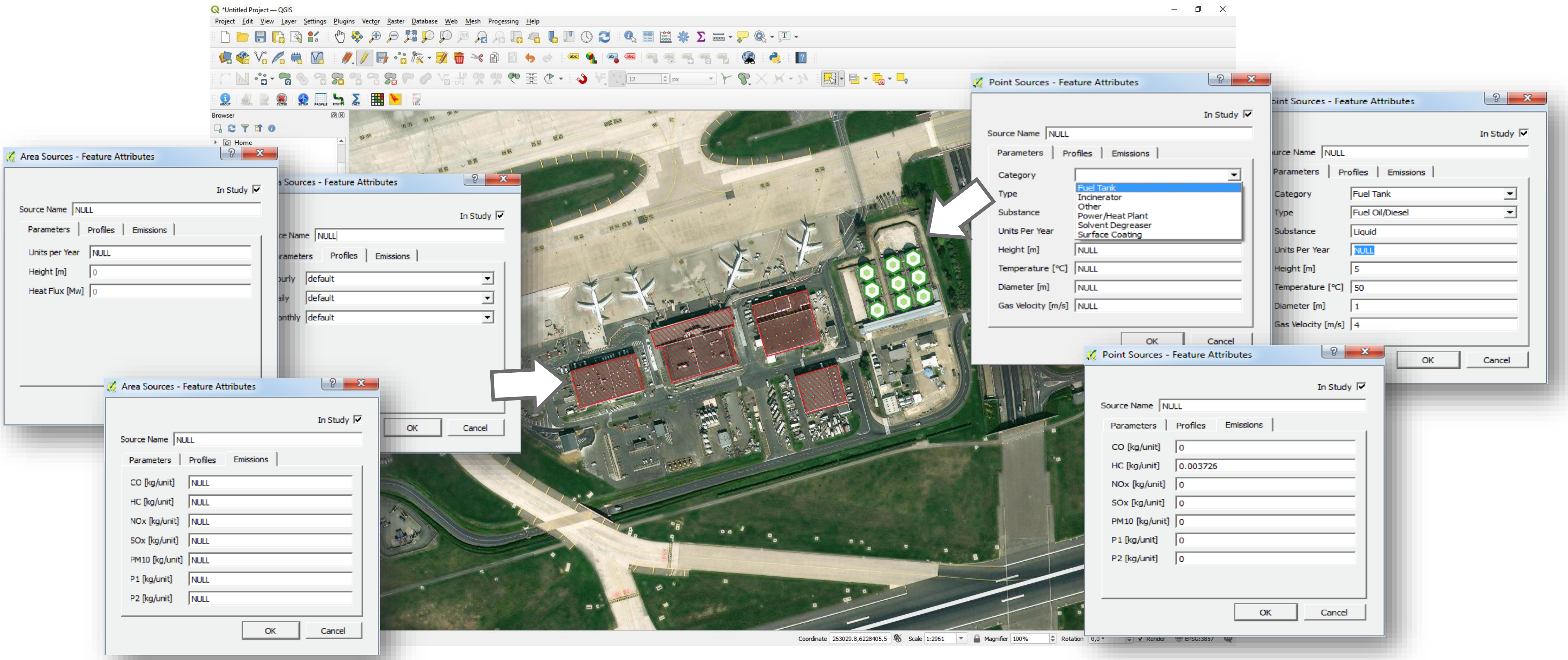
Edit Route

Select Taxiway Segments

Select Aircraft Groups



Stationary Sources



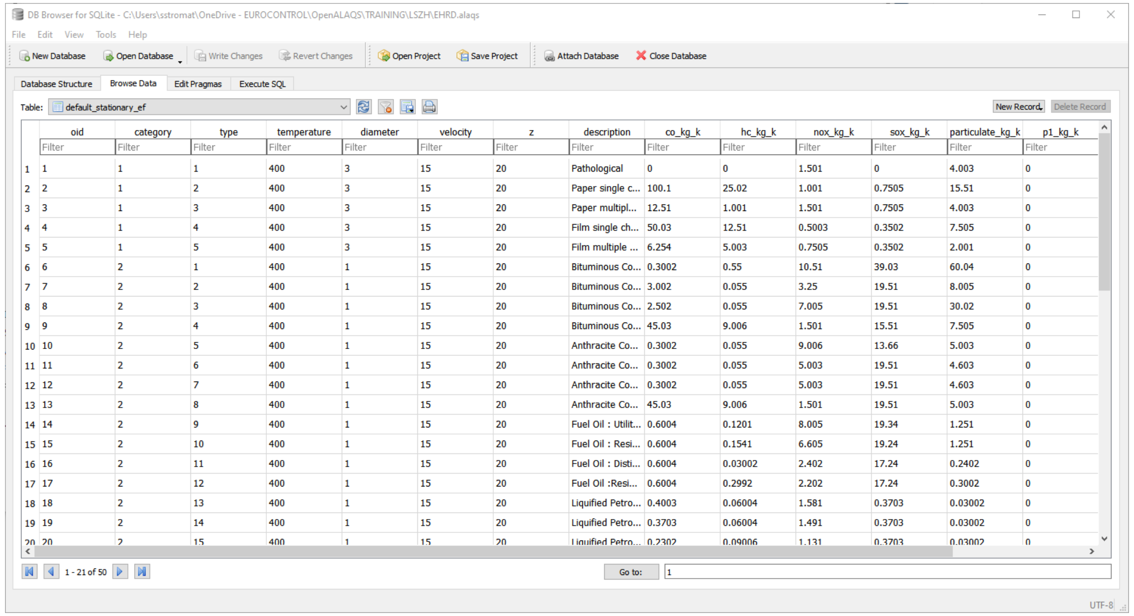
Default or custom EFs for each source

Custom (user-defined) EFs for each source

Stationary Sources

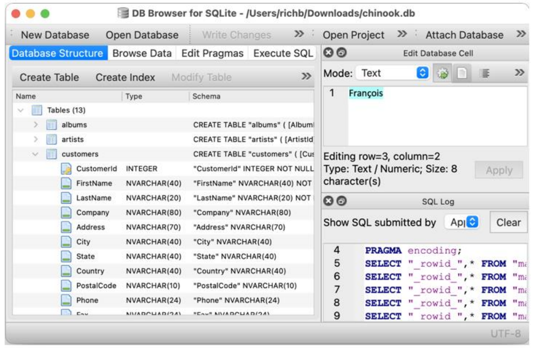
Create 1 point source and 1 area source

Modify the default emission factors for the point source category and type



DB Browser for SQLite

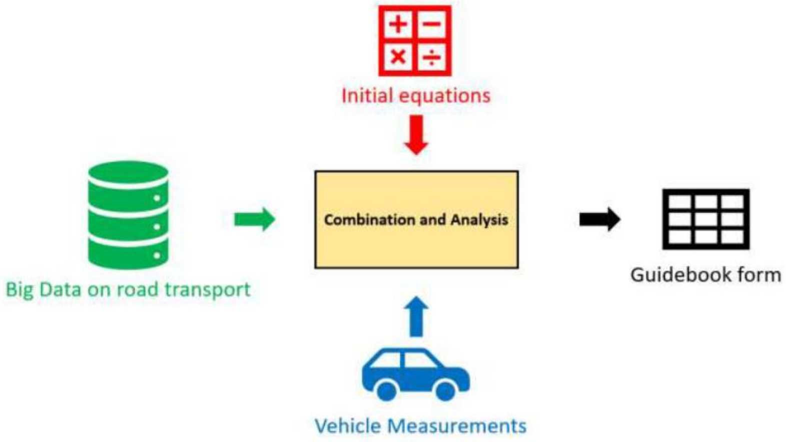
DB Browser for SQLite (DB4S) is a high quality, visual, open source tool designed for people who want to create, search, and edit SQLite or SQLCipher database files. DB4S gives a familiar spreadsheet-like interface on the database in addition to providing a full SQL query facility. It works with Windows, macOS, and most versions of Linux and Unix. Documentation for the program is on the wiki.



OpenALAQS files can be easily edited with DB Browser for SQLite an open-source tool designed for manipulating SQLite database files.

Roadway traffic

- Estimation of **Roadway traffic emissions** (landside, airside, parking lots) is based on **COPERT** (version V)
- COPERT contains **emission factors for more than 450 individual vehicle types** (e.g. PC, LDV, HDV)
- Estimates **emissions from all relevant road vehicle operation modes**:
  - ‘hot’ & ‘cold start’ emissions
  - non-exhaust emissions (from fuel evaporation, tyre and brake wear emissions)
- The development of COPERT is **coordinated by the European Environment Agency** (EEA)



https://www.emisia.com

Roadway traffic

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Parking Lots

Emissions from parking areas for vehicles are estimated based on the `COPERT` methodology.

When adding an airport parking lot, the following information is required:

- Parameters
  - **Number per year**: Total number of vehicles per year
  - **Height**: Height at which emissions are released (in meters) `not yet fully implemented`
  - **Speed**: Average travel speed in parking (in km/h)
  - **Travel distance**: Average travel distance in parking (in meters)
  - **Idle time**: Vehicle average idling time between entry and exit (in minutes)
  - **Average parking time**: Average time a vehicle remains on parking (in minutes)
- Profiles
  - Hourly, Daily or Monthly activity profiles
- Fleet mix
  - **PC (Petrol) [in %]**: Passenger Cars (Petrol)
  - **PC (Diesel) [in %]**: Passenger Cars (Diesel)
  - **LDV (Petrol) [in %]**: Light Duty Vehicles (Petrol)
  - **LDV (Diesel) [in %]**: Light Duty Vehicles (Diesel)
  - **HDV (Petrol) [in %]**: Heavy Duty Vehicles (Petrol)
  - **HDV (Diesel) [in %]**: Heavy Duty Vehicles (Diesel)
  - **Motorcycles [in %]**
  - **Buses [in %]**

Roadways

Airside or landside emissions are calculated using the same methodology as described above.

When adding a roadway, the following information is required:

- Parameters
  - **Movements per year**: Number of annual movements
  - **Height**: Height at which emissions are released (in meters) `not yet fully implemented`
  - **Speed**: Vehicles speed in roadway (in km/h)
- Profiles:
  - Hourly, Daily or Monthly activity profiles
- Fleet mix
  - **PC (Petrol) [in %]**: Passenger Cars (Petrol)
  - **PC (Diesel) [in %]**: Passenger Cars (Diesel)
  - **LDV (Petrol) [in %]**: Light Duty Vehicles (Petrol)
  - **LDV (Diesel) [in %]**: Light Duty Vehicles (Diesel)
  - **HDV (Petrol) [in %]**: Heavy Duty Vehicles (Petrol)
  - **HDV (Diesel) [in %]**: Heavy Duty Vehicles (Diesel)
  - **Motorcycles [in %]**
  - **Buses [in %]**

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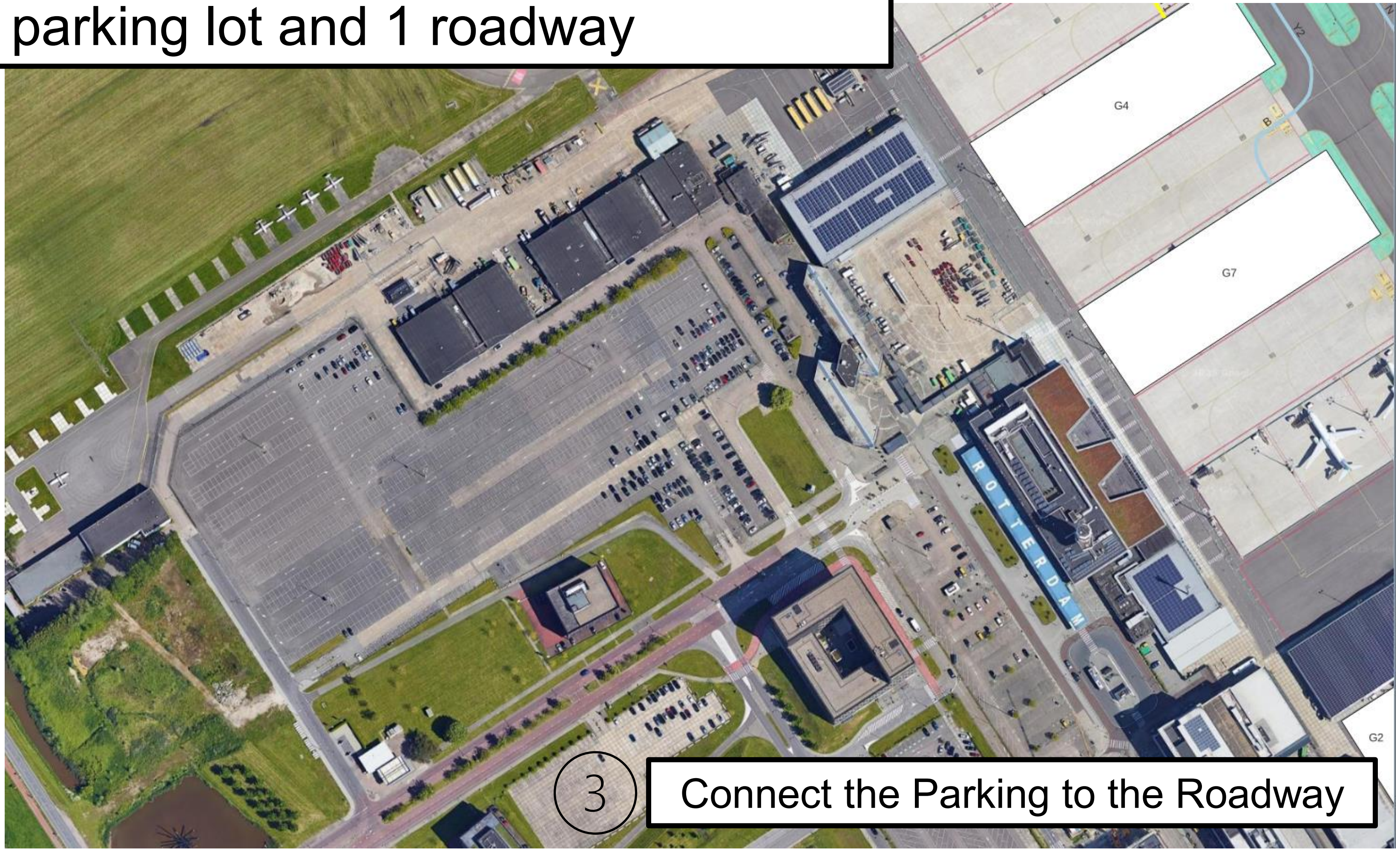
Roadway traffic

Create 1 parking lot and 1 roadway

1. **Create Parking Lot**
   - *Provide values for all necessary parameters*
   - *Calculate EFs*

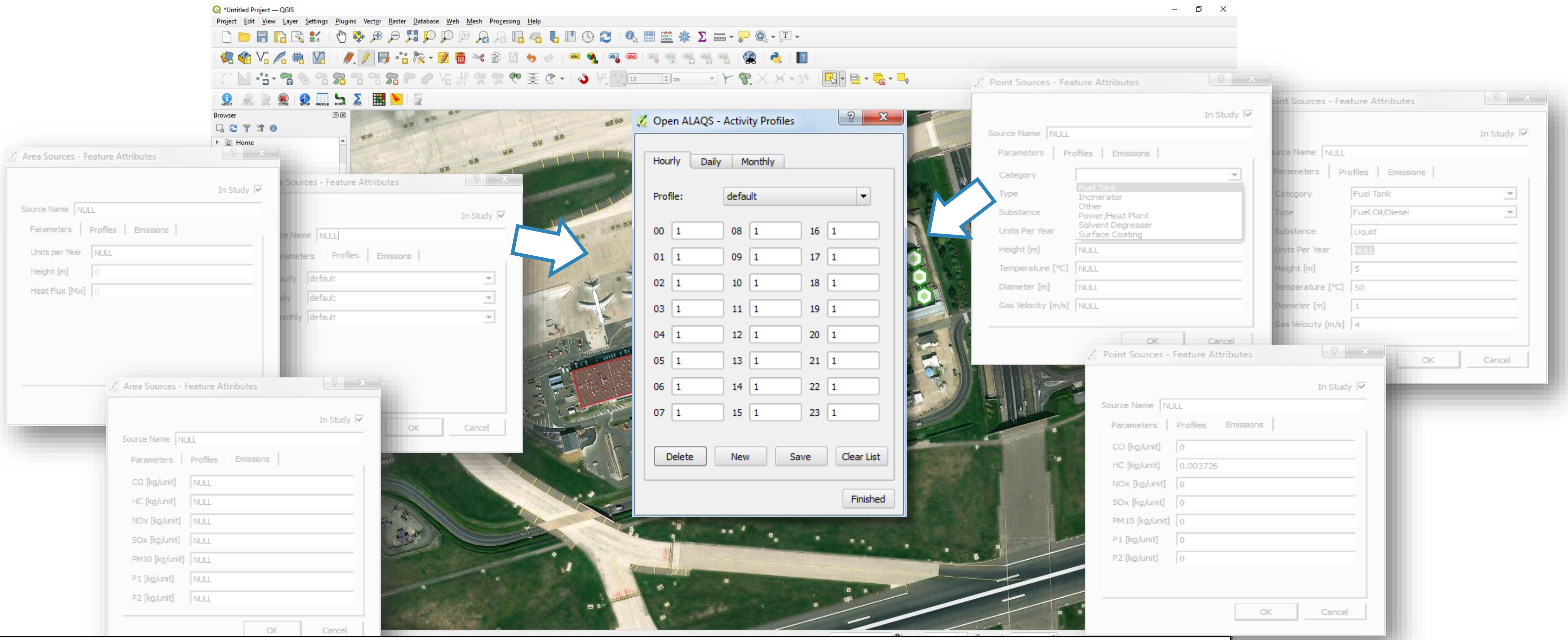
2. **Create Roadway**
   - *Provide values for all necessary parameters*
   - *Calculate EFs*



3. Connect the Parking to the Roadway

Activity Profiles

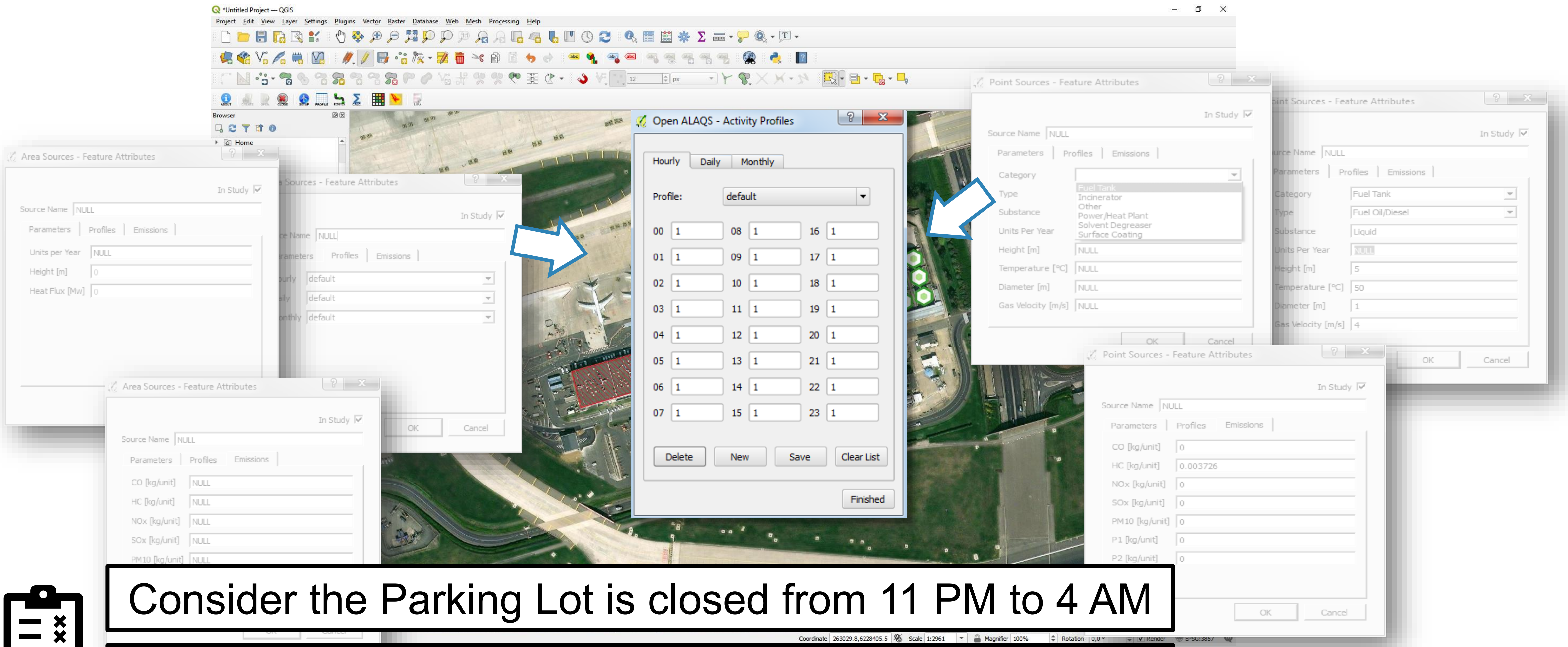
(Optional)



Each activity multiplier is a decimal number, between 0 and 1.
The default profile values are 1 (i.e., 100%) meaning the emission source is fully active.

Activity Profiles

(Optional)



Consider the Parking Lot is closed from 11 PM to 4 AM

Create an Activity Profile and apply it to the Parking Lot

Emission Sources

Generate Emissions Inventory – Input Data

Aircraft movements

1

runway_time	block_time	aircraft_registration	aircraft	gate	departure_arrival	runway	engine_name	prof_id	track_id	taxi_route	tow_ratio	apu_code	taxi_engine_count
2015-01-01 05:18:00	2015-01-01 05:23:00		B744	T01_T19	A	24	1PW042	747400-A-1			1		1
2015-01-01 06:27:00	2015-01-01 06:31:30		MD11	T01_T19	A	24	1PW059	MD11GE-A-1			1		1
2015-01-01 07:20:00	2015-01-01 07:25:00		MD11	T01_T19	A	24	1PW059	MD11GE-A-1			1		1
2015-01-01 08:17:00	2015-01-01 08:08:30		MD11	T01_T19	D	24	1PW059	MD11GE-D-7			1		1
2015-01-01 09:21:38	2015-01-01 09:25:21		T154	T01_T19	A	24	1KK002	727D17-A-1			1		1
2015-01-01 09:39:00	2015-01-01 09:27:00		MD11	T01_T19	D	24	1PW059	MD11GE-D-7			1		1
2015-01-01 10:30:16	2015-01-01 10:34:18		E145	T01_T19	A	24	6AL006	CL601-A-1			1		1
2015-01-01 10:44:08	2015-01-01 10:38:40		T154	T01_T19	D	24	1KK002	727D17-D-3			1		1
2015-01-01 11:02:29	2015-01-01 11:07:12		A319	T20_T29	A	24	3CM028	A319-A-1			1		1
2015-01-01 11:52:00	2015-01-01 11:43:30		MD11	T01_T19	D	24	1PW059	MD11GE-D-7			1		1

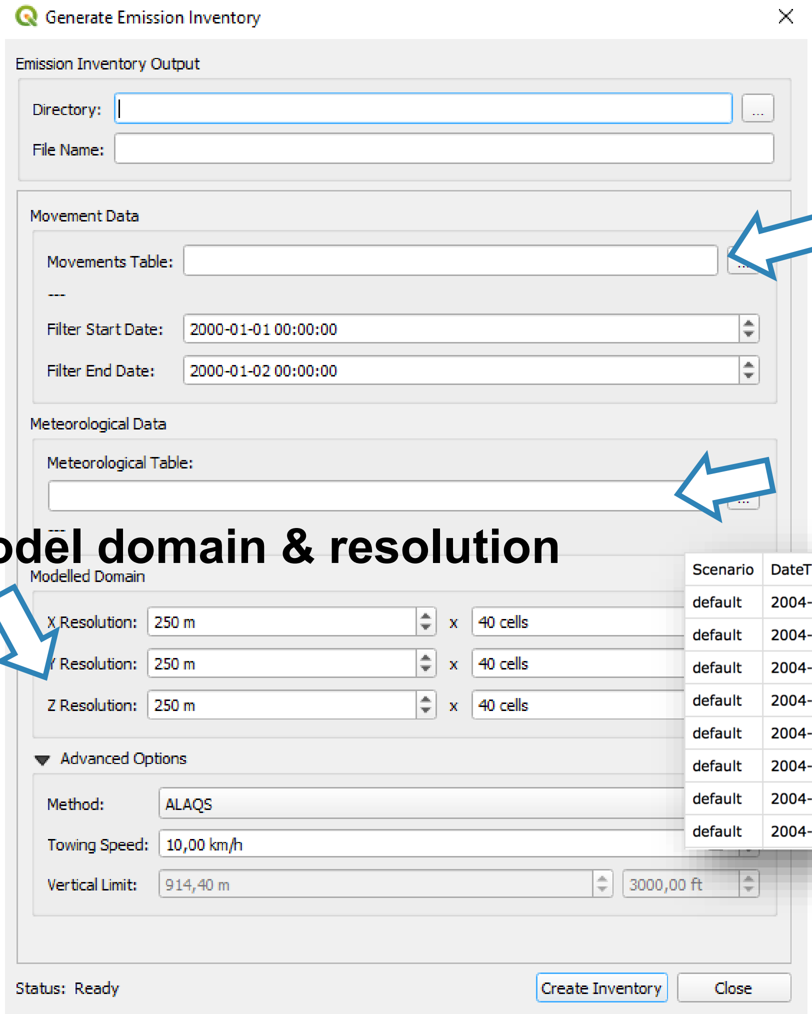
Meteorology

2

Scenario	DateTime(YYYY-mm-dd hh:mm:ss)	Temperature(K)	Humidity(kg_water/kg_dry_air)	RelativeHumidity(%)	SeaLevelPressure(mb)	WindSpeed(m/s)	WindDirection(degrees)	ObukhovLength(m)	MixingHeight(m)
default	2004-01-01 00:00:00	276.48	0.00436	0.93	100960	0.89	280	-99999	914.4
default	2004-01-01 06:00:00	275.93	0.00406	0.93	100960	0.89	140	6.2	914.4
default	2004-01-01 12:00:00	279.82	0.0047	0.81	100540	6.26	180	-8888	914.4
default	2004-01-01 18:00:00	279.82	0.00588	1	99270	9.83	180	1954.1	914.4
default	2004-01-02 00:00:00	281.48	0.00551	0.81	98740	6.26	220	436.9	914.4
default	2004-01-02 06:00:00	282.59	0.00637	0.87	98110	5.81	180	647.4	914.4
default	2004-01-02 12:00:00	280.93	0.00519	0.81	97710	4.02	240	-5268.3	914.4
default	2004-01-02 18:00:00	281.48	0.00476	0.71	99280	4.02	250	-99999	914.4

Model domain & resolution

3



Movements Table

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This table contains all the aircraft movements occurring at the airport during a certain period. It must include the following information:

- **runway_time**: Date and time of arrival at the airport runway (YYYY-MM-DD HH:MM:SS)
- **block_time**: Date and time of arrival at the gate (YYYY-MM-DD HH:MM:SS)
- **aircraft**: The ICAO aircraft type
- **gate**: The stand used for the aircraft operation
- **departure_arrival**: The type of operation (arrival or departure)
- **runway**: The name of the runway used for the aircraft operation
- **taxi_route**: the name of the taxi route used for the aircraft operation
- **apu_code**: code defining APU usage away from gate/stand (0: APU used at gate only, 1: APU on while taxiing, 2: APU on during taxiing, take-off, climb or approach)

The rest of the parameters are not mandatory and can be left empty. They will only be used if the user provides specific values. Otherwise, the default values from the internal database will be used.

- **aircraft_registration**: aircraft registration number `not yet fully implemented`
- **engine_name**: engine name identifier (from the internal database)
- **profile_id**: performance profile identifier (from the internal database)
- **track_id**: aircraft trajectory identifier `not yet fully implemented`
- **tow_ratio**: take-off gross weight divided by maximum take-off weight (float, <=1) `not yet fully implemented`
- **taxi_engine_count**: number of engines used during taxing (integer)
- **set_time_of_main_engine_start_after_block_off_in_s**: time spent for MES after leaving the gate (in seconds)
- **set_time_of_main_engine_start_before_takeoff_in_s**: time spent for MES before take-off (in seconds)
- **set_time_of_main_engine_off_after_runway_exit_in_s**: time spent for MES after leaving the runway exit (in seconds)
- **engine_thrust_level_for_taxiing**: taxing thrust level setting (ICAO default: 7%) `not yet fully implemented`
- **taxi_fuel_ratio**: ratio between actual fuel flow and fuel for the idle mode (float) `not yet fully implemented`
- **number_of_stop_and_gos**: number of stop and go points during taxing `not yet fully implemented`
- **domestic**: domestic/international flight (Y or N) `not yet fully implemented`

(Optional)

Movements Table

EUROCONTROL

This table contains all the aircraft movements occurring at the airport during a certain period. It must include the following information:

- **runway_time**: Date and time of arrival at the airport runway (YYYY-MM-DD HH:MM:SS)

Use the provided example file to create a Movements Table .csv file for your study

Create at least 1 ARR and 1 DEP for each gate and runway direction

hile taxiing, 2: APU on during taxiing,

provides specific values. Otherwise, the default values from the internal database will be used.

Consider different AC types

- **profile_id**: performance profile identifier (from the internal database)
- **track_id**: aircraft trajectory identifier `not yet fully implemented`
- **tow_ratio**: take-off gross weight divided by maximum take-off weight (float, <=1) `not yet fully implemented`
- **taxi_engine_count**: number of engines used during taxing (integer)
- **set_time_of_main_engine_start_after_block_off_in_s**: time spent for MES after leaving the gate (in seconds)
- **set_time_of_main_engine_start_before_takeoff_in_s**: time spent for MES before take-off (in seconds)
- **set_time_of_main_engine_off_after_runway_exit_in_s**: time spent for MES after leaving the runway exit (in seconds)
- **engine_thrust_level_for_taxiing**: taxing thrust level setting (ICAO default: 7%) `not yet fully implemented`
- **taxi_fuel_ratio**: ratio between actual fuel flow and fuel for the idle mode (float) `not yet fully implemented`
- **number_of_stop_and_gos**: number of stop and go points during taxing `not yet fully implemented`
- **domestic**: domestic/international flight (Y or N) `not yet fully implemented`

(Optional)

Meteorological Data

EUROCONTROL

OpenALAQS requires certain meteorological data in order to accurately calculate emissions using advanced methods (e.g. BFFM2 method, correction of NOx for ambient conditions) but also for the dispersion calculation.

The required meteorological parameters are:

- **Scenario**: simulation name identifier
- **DateTime** (in YYYY-MM-DD HH:MM:SS)
- **Temperature** (in K)
- **Humidity** (in kg water/kg dry air)
- **RelativeHumidity** (in %)
- **SeaLevelPressure** (in Pa)

open alaqs

- **WindSpeed** (in m/s)
- **WindDirection** (in degrees)
- **ObukhovLength** (in m)
- **MixingHeight** (in m)

AUSTAL

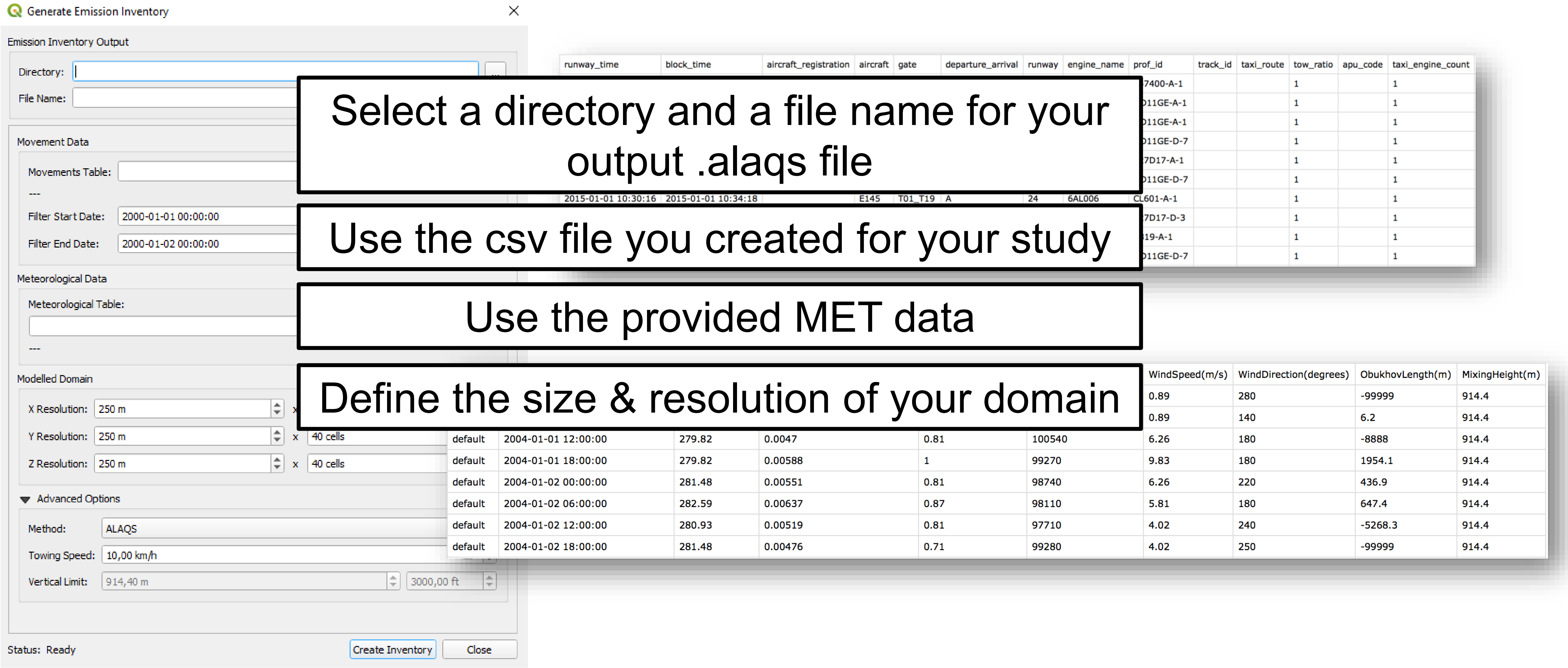
Input data may come from local or national data providers (e.g. METAR, SYNOP) or reanalysis data. The compilation of the meteorological data file for OpenALAQS/AUSTAL2000 is left to the user. Otherwise, default ISA conditions values are considered for all necessary meteorological parameters.

Climate Data Store

PROGRAMME OF THE EUROPEAN UNION

https://github.com/ecmwf/cdsapi

Generate Emissions Inventory – Input Data



Select a directory and a file name for your output .alaqs file

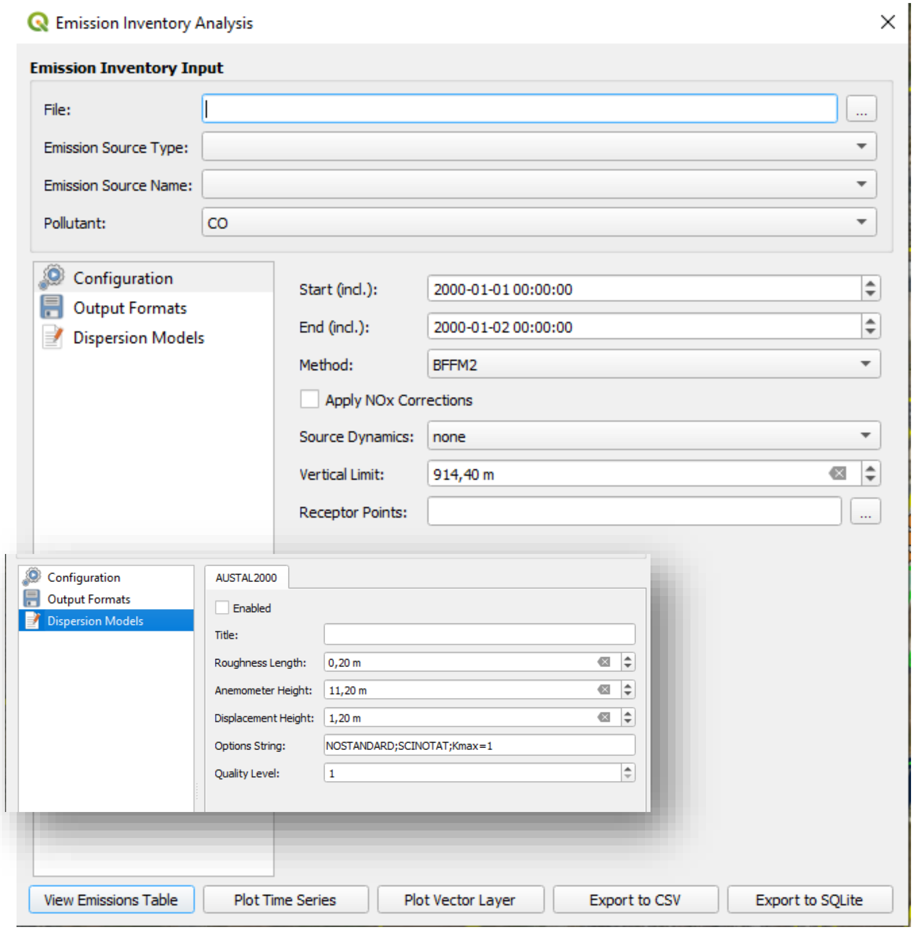
Use the csv file you created for your study

Use the provided MET data

Define the size & resolution of your domain

Calculate and Visualise Emissions

- **Calculate emissions:**
  - Per Source Type
  - Per Source Name
  - Per Pollutant
- Dispersion module can be activated/deactivated and configured from within Open-ALAQS
- Various ways of visualising/exporting the results
  - Emissions are aggregated and presented for each hour of the simulation time period

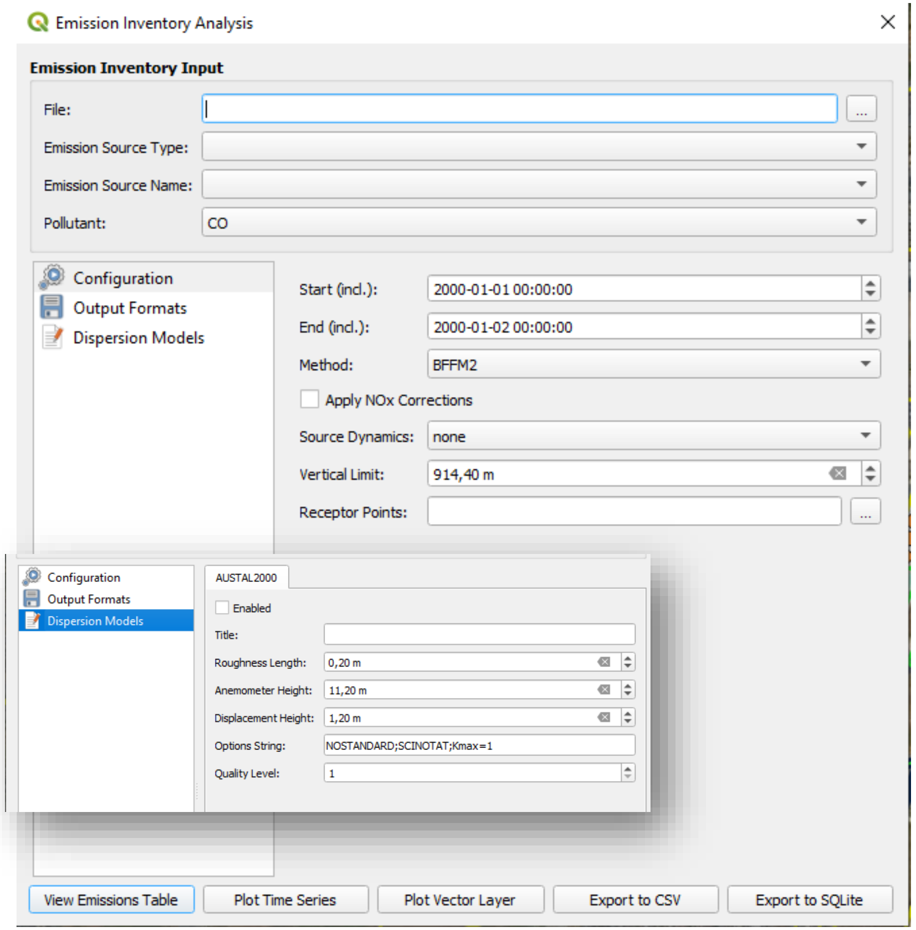


Calculate and Visualise Emissions

Select the output .alaqs file you've created before

Calculate the Parking Lot CO emissions

Visualise the calculated emissions as a Table and Vector Layer



Aircraft emissions

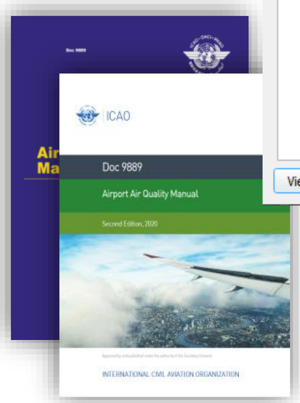
- **Different options for calculating aircraft emissions**:
  - By ICAO mode (TX, TO, CL, AP)
  - Boeing Fuel Flow Method (BFFM2)

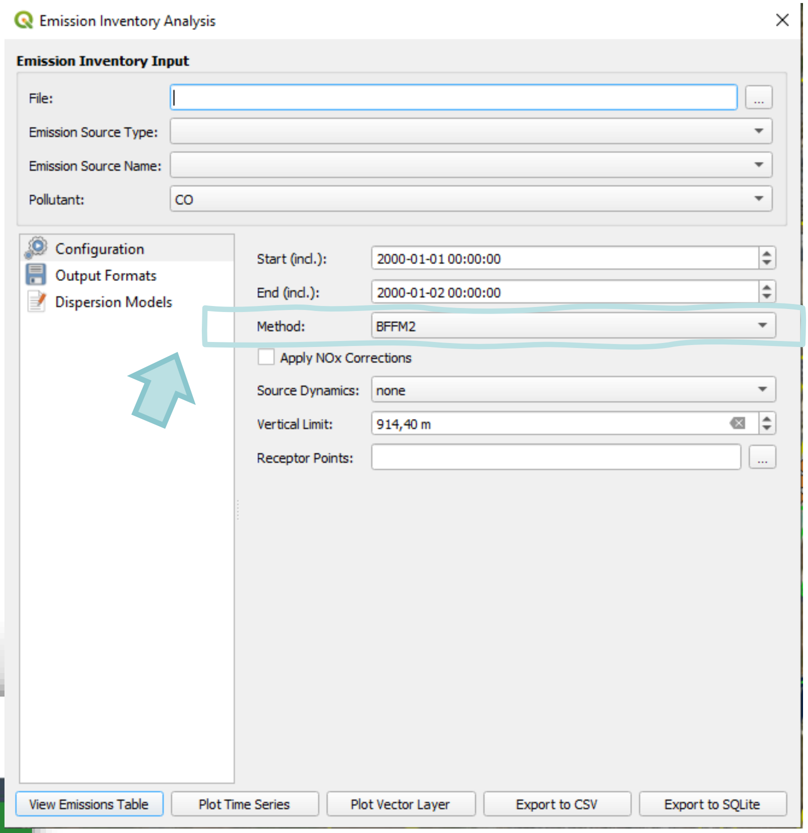
- **Emission Factors:**
  - **ICAO Engine Emissions Databank** for aircraft jet engine
  - **FOCA** Piston Engine EFs
  - Possible to use **turboprops** EFs from the **FOI** data base

- **Aircrafr Performance:**
  - Aircraft Noise and Performance (ANP) Database

ICAO ENVIRONMENT

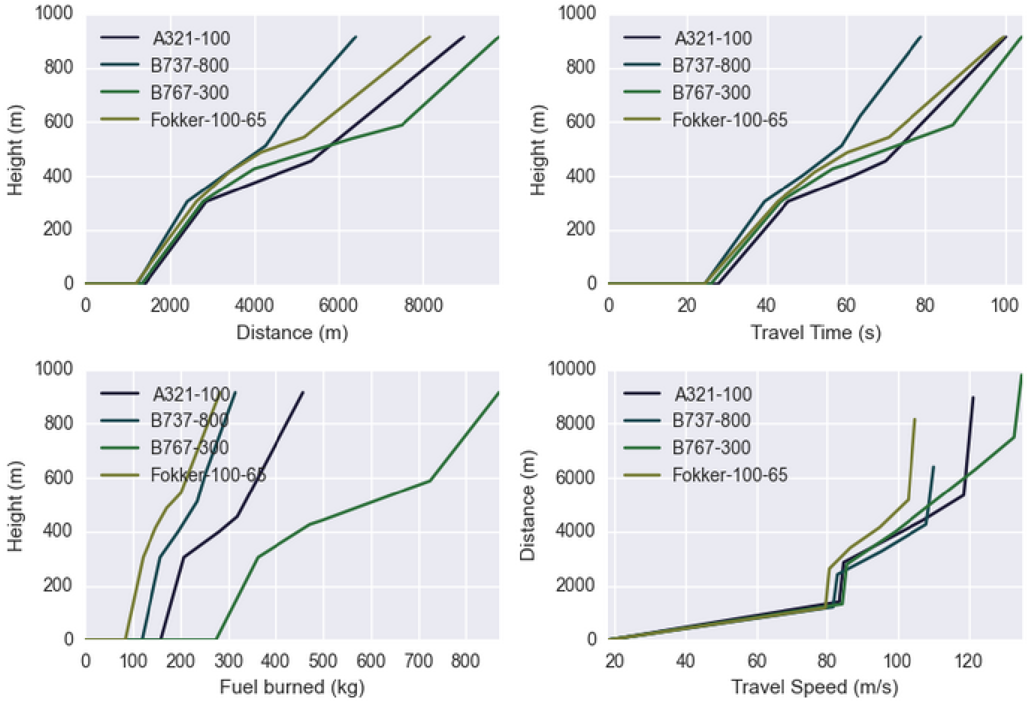
CAEP - **Modelling and Databases Group (MDG)**





Aircraft Noise and Performance (ANP) Data

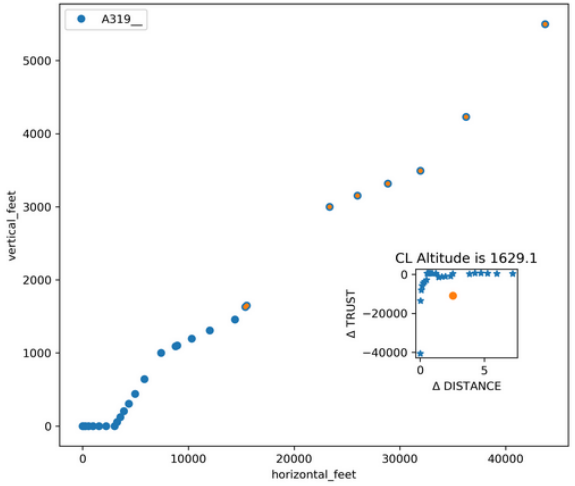
In OpenALAQS, the ANP fixed-point profiles (see `default_aircraft_profiles`) are used to calculate aircraft emissions. The ANP fixed-point profiles contain information on the relative 2D trajectory of the aircraft (horizontal versus vertical distance) on the runway. These points are converted into an aircraft trajectory for a given runway based on its geographic coordinates. Currently, only straight-line trajectories are possible in OpenALAQS.



Performance profiles

The ratio of thrust to distance is used to define the cut-off between take-off and climb-out. During take-off, full thrust is required to accelerate the aircraft. As the aircraft reaches a certain distance and speed, thrust is reduced to a level appropriate for climb. This transition involves reducing thrust from maximum take-off to maximum climb thrust after a set distance, typically around 1000 feet of ground distance. This cut-off point is used in OpenALAQS to separate the two modes.

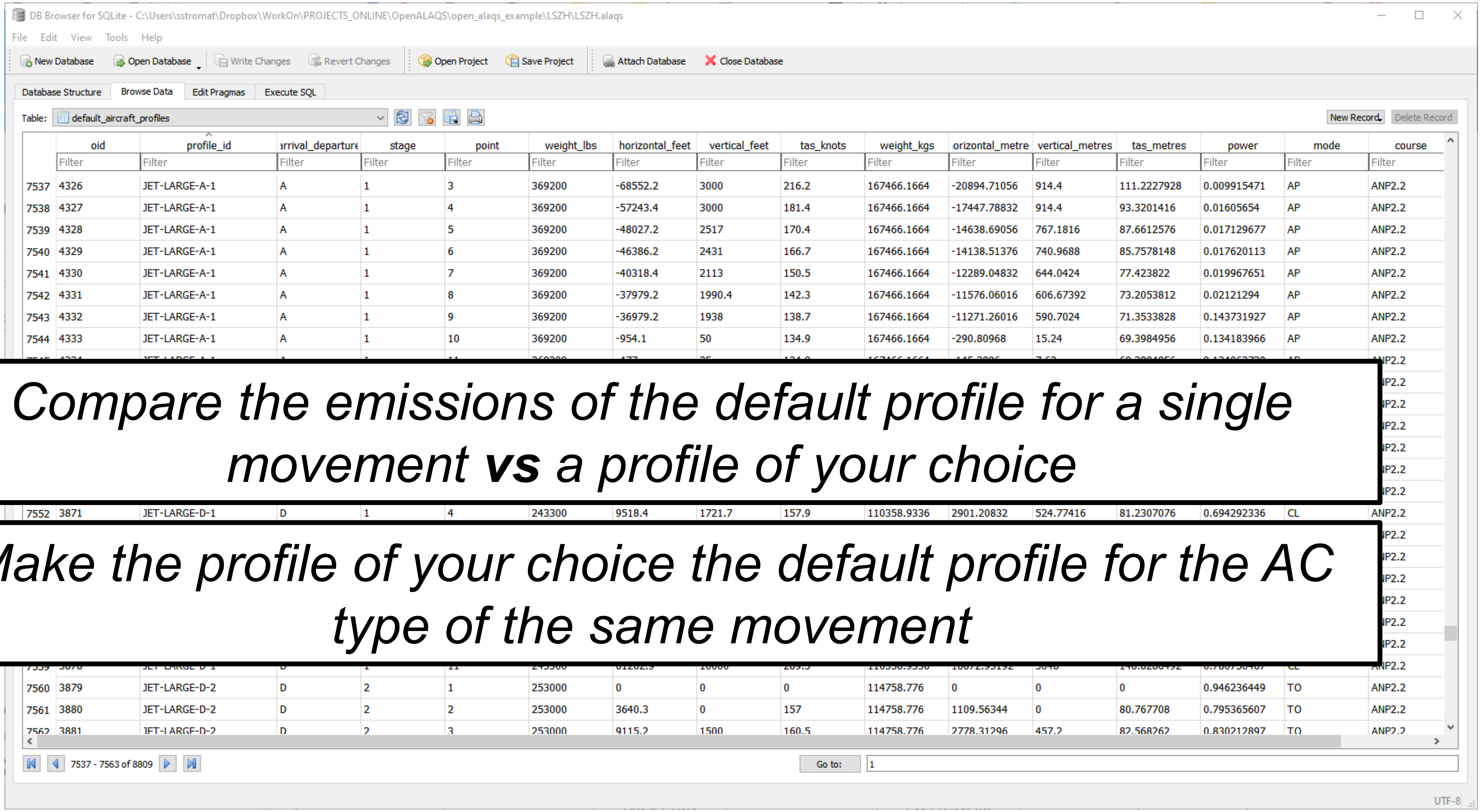
The following figure illustrates this approach. For more information the user is referred to `ECAC.CEAC Doc 29, Volume 2, Appendix B`.



Aircraft Noise and Performance (ANP) Data

https://www.easa.europa.eu/en/domains/environment/policy-support-and-research/aircraft-noise-and-performance-anp-data

Aircraft Noise and Performance (ANP) Data



Compare the emissions of the default profile for a single movement **vs** a profile of your choice

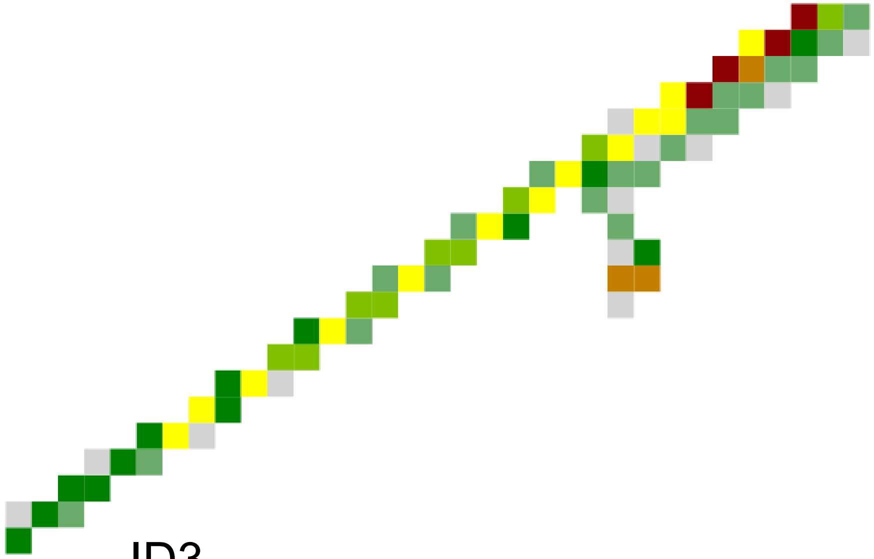
Make the profile of your choice the default profile for the AC type of the same movement

Aircraft Noise and Performance (ANP) Data

EUROCONTROL

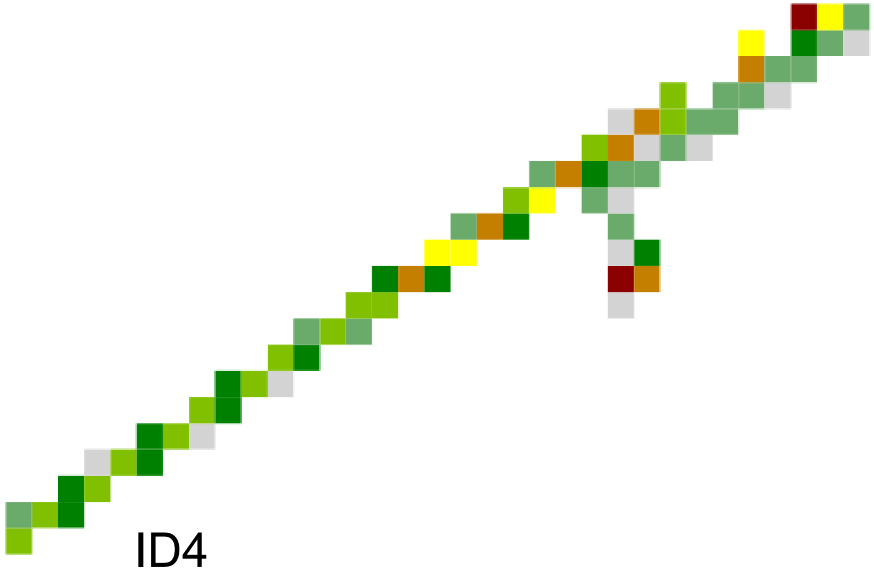
*Compare the emissions of the default profile for a single movement **vs** a profile of your choice*

3	3	2025-12-01 06:40:00	2025-12-01 06:35:00		A21N	G2	D	06	01P18PW157	JET-SMALL-D-1		G2/24/D/1
4	4	2025-12-01 06:41:00	2025-12-01 06:36:00		A21N	G2	D	06	01P18PW157	A320-232-D-1		G2/24/D/1



ID3

JET-SMALL-D-1



ID4

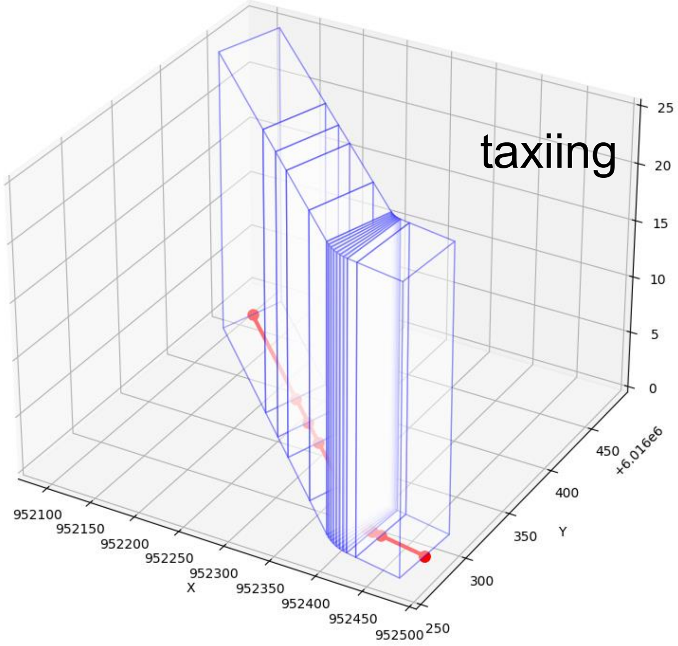
A320-232-D-1

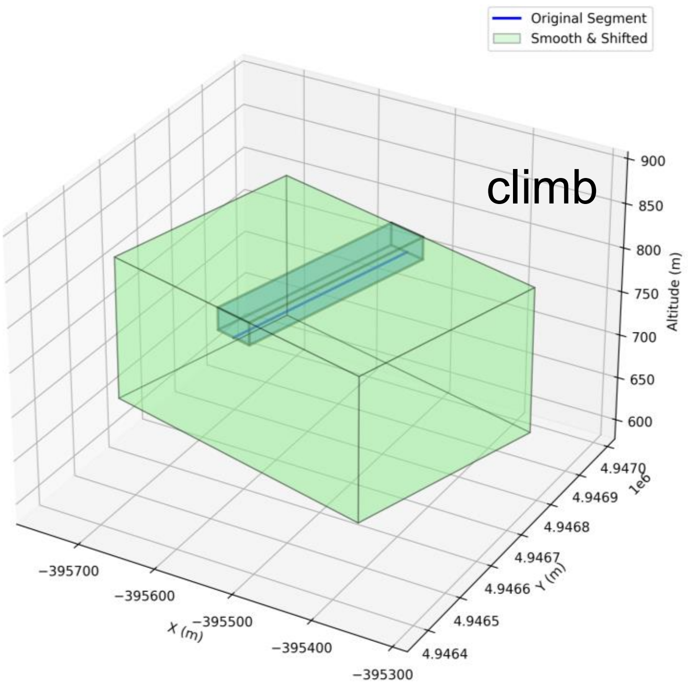
nox Emissions - ID4
- 0
- 0 - 0,0101
- 0,0101 - 0,0301
- 0,0301 - 0,0593
- 0,0593 - 0,0895
- 0,0895 - 0,1183
- 0,1183 - 0,1767
- 0,1767 - 0,2399

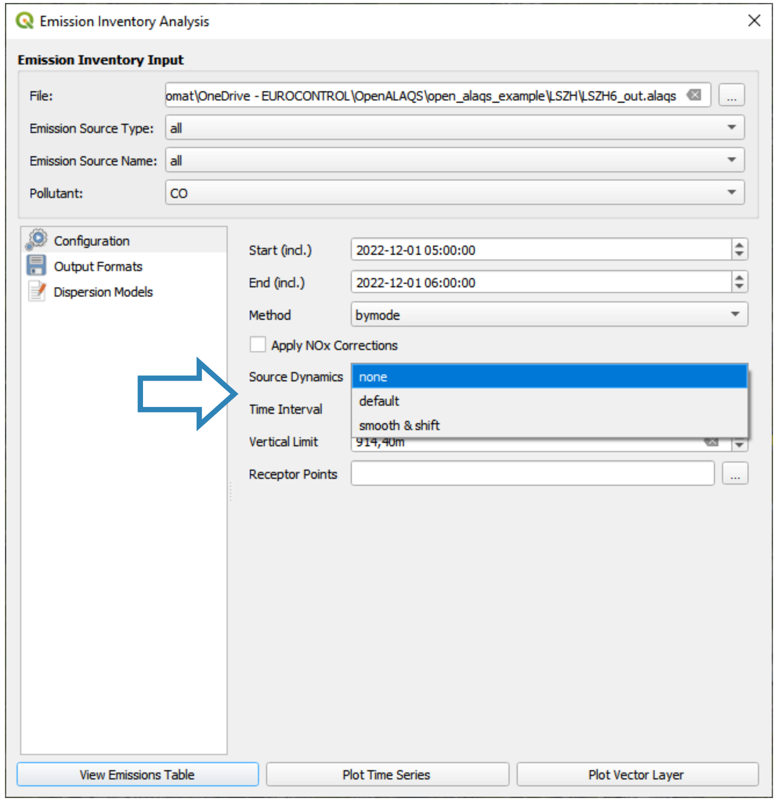
nox Emissions - ID3
- 0
- 0 - 0,0101
- 0,0101 - 0,0301
- 0,0301 - 0,0593
- 0,0593 - 0,0895
- 0,0895 - 0,1183
- 0,1183 - 0,1767
- 0,1767 - 0,2399

Exhaust Dynamics (Smooth and Shift)

Possible to account for source dynamics such as turbulence, exhaust momentum from aircraft engines, and thermal plume rise.

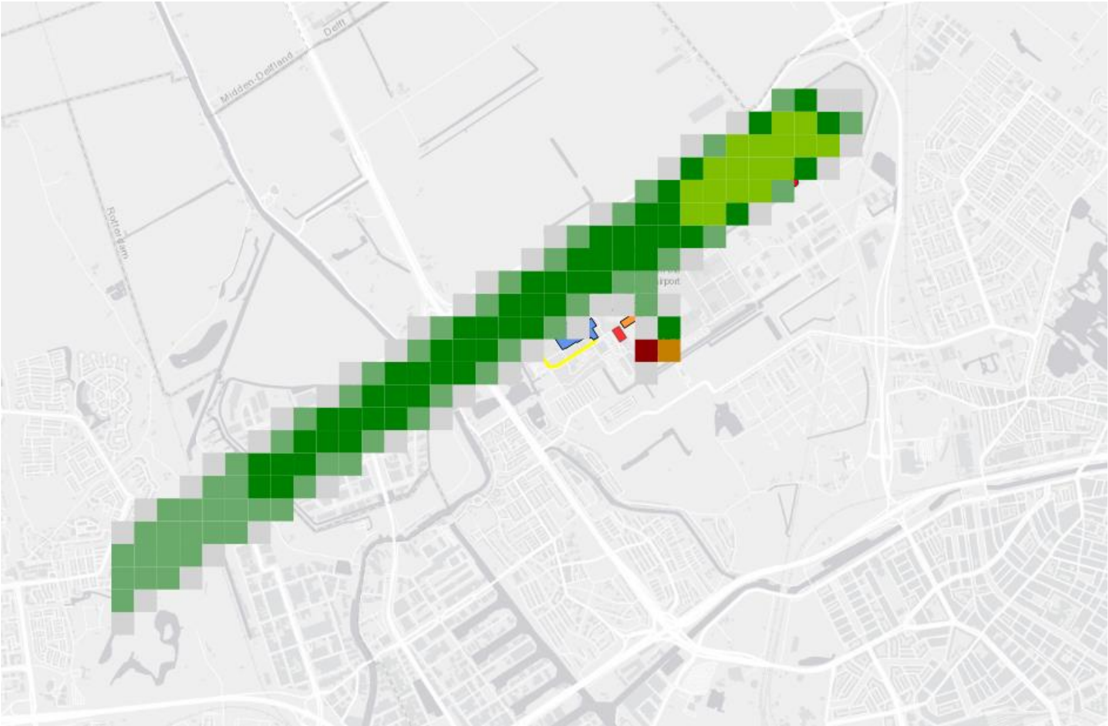






Exhaust Dynamics (Smooth and Shift)

Visualise the emissions of a single movement using the available options



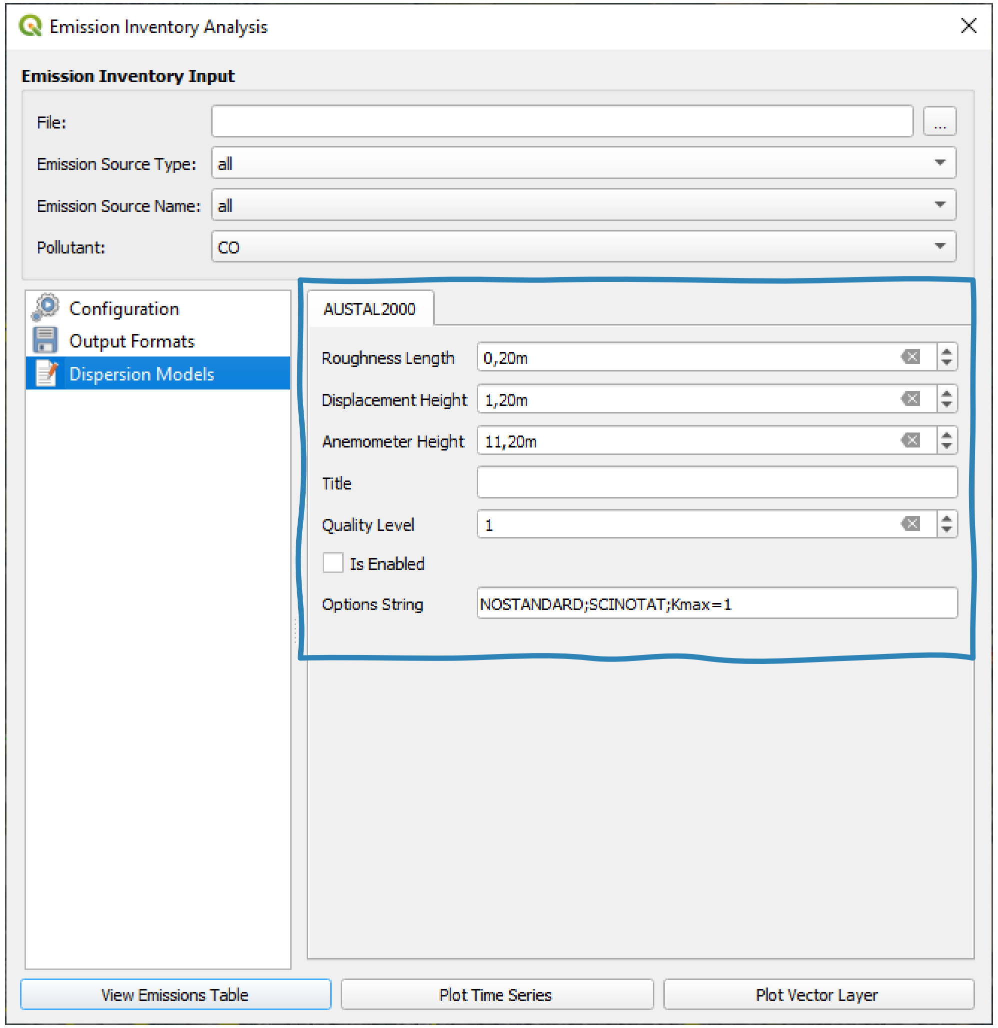
Dispersion modelling AUSTAL

- **Lagrangian particle model** in conformance with the German guideline VDI 3945 Part 3.
- The official reference model of the German Regulation on Air Quality Control (*Technische Anleitung zur Reinhaltung der Luft*, TA Luft)
- It **accounts for several different pollutants**, each with specific results, which are strictly bound to the regulations of the German Standard TA Luft 2002 (acc. to European EC Directive 2008/50/EC)
- AUSTAL (v3.3.0) is **provided free of charge** under the GNU Public Licence

https://www.umweltbundesamt.de/en/topics/air/air-quality-control-in-europe/download

AUSTAL Parameters:

- **Roughness Length** (*set to 0.2m by default*)
- **Displacement Height** (*set to 1.2m by default*)
- **Anemometer Height** (*set to 11.2m by default*)
- **Title**: The project name
- **Quality Level** (*set to 1 by default*)
- **Is Enabled**: Enable or disable dispersion module
- **Options String**: Define advanced AUSTAL settings



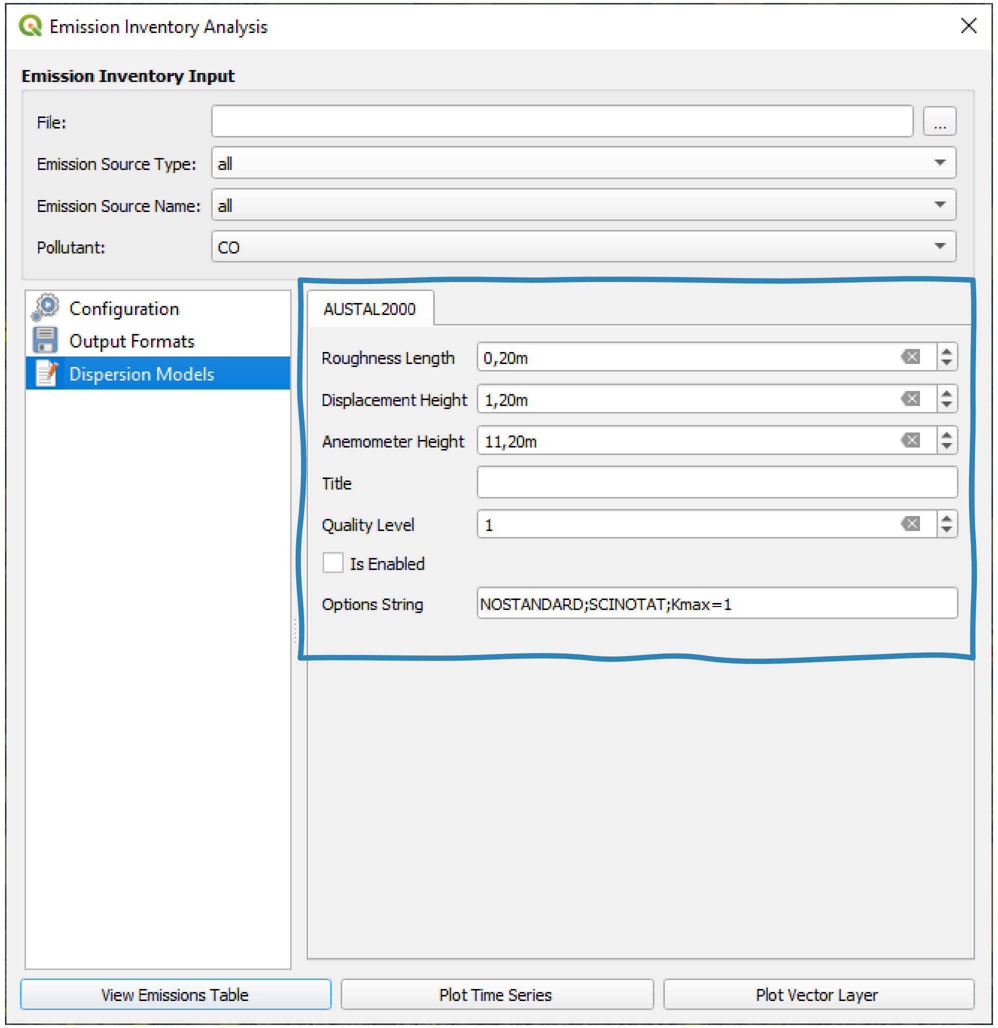
https://github.com/eurocontrol-asu/open_alaqs/blob/main/open_alaqs/assets/emissions-calculation-2.png#dispersion-modeling-with-austal

AUSTAL Parameters:

- **Roughness Length** (*set to 0.2m by default*)
- **Displacement Height** (*set to 1.2m by default*)
- **Anemometer Height** (*set to 11.2m by default*)
- **Title**: The project name
- **Quality Level** (*set to 1 by default*)
- **Is Enabled**: Enable or disable dispersion module
- **Options String**: Define advanced AUSTAL settings

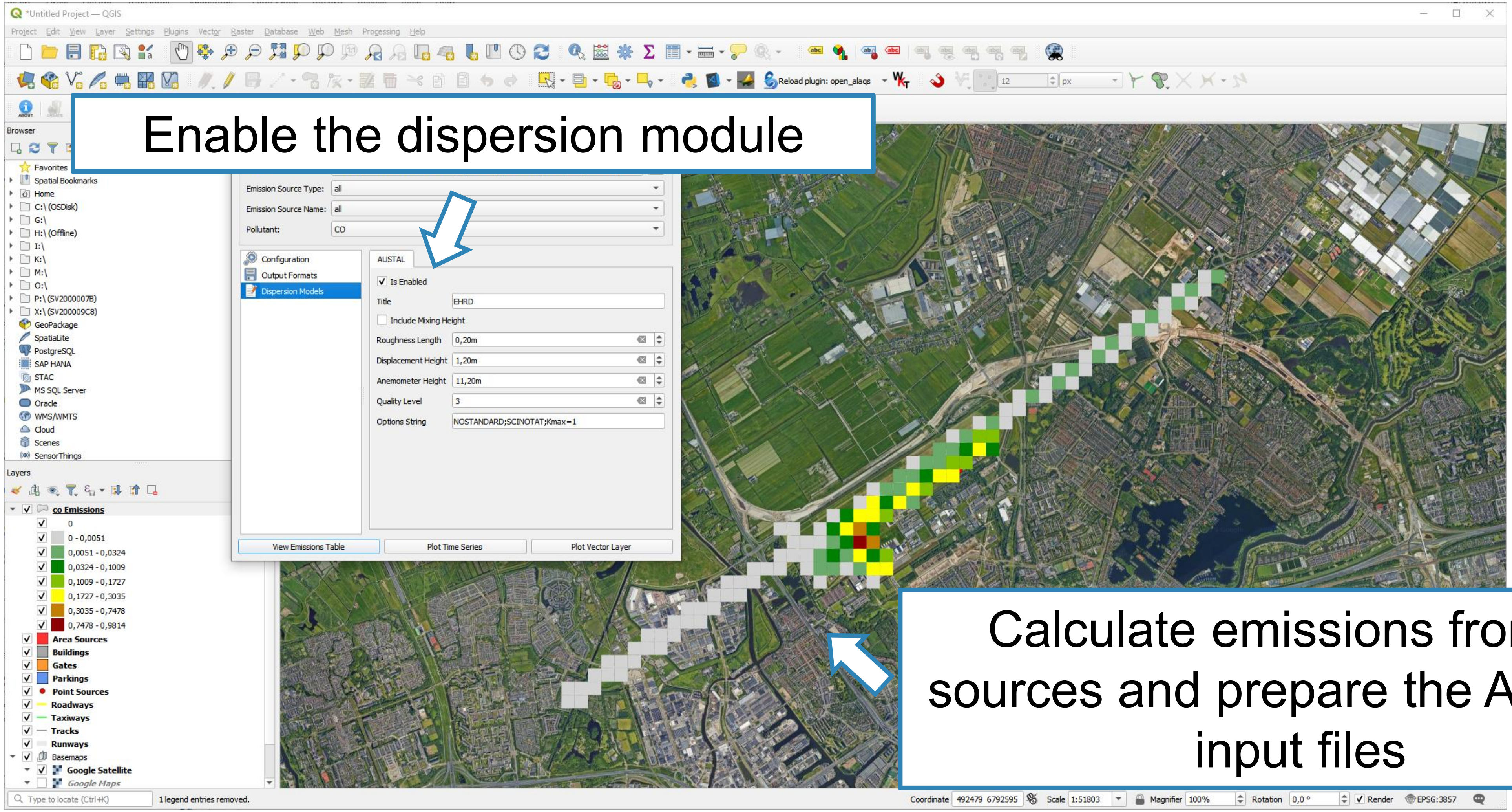
Enable the dispersion module

Calculate emissions from all sources and prepare the AUSTAL input files



https://github.com/eurocontrol-asu/open_alaqs/blob/main/open_alaqs/assets/emissions-calculation-2.png#dispersion-modeling-with-austal

AUSTAL Parameters:



Enable the dispersion module

Calculate emissions from all sources and prepare the AUSTAL input files

Running AUSTAL

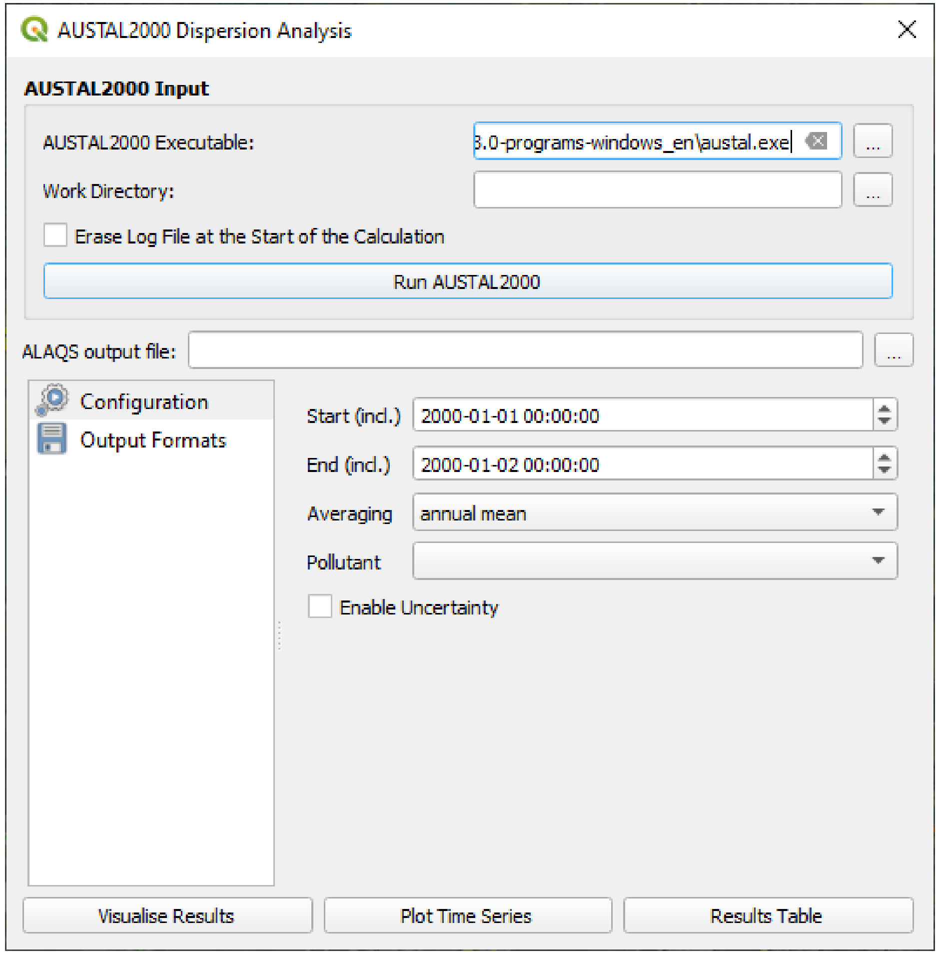
Specify the path to the **AUSTAL executable** and the project directory (**Work Directory**) where all output files will be written.

Run AUSTAL to start the dispersion calculation.

Select the .alaqs file used for generating the AUSTAL input data

Select the pollutant used and the averaging interval

Visualized the calculated concentrations on a map



https://github.com/eurocontrol-asu/open_alaqs/blob/main/open_alaqs/assets/emissions-calculation-2.png#dispersion-modeling-with-austal

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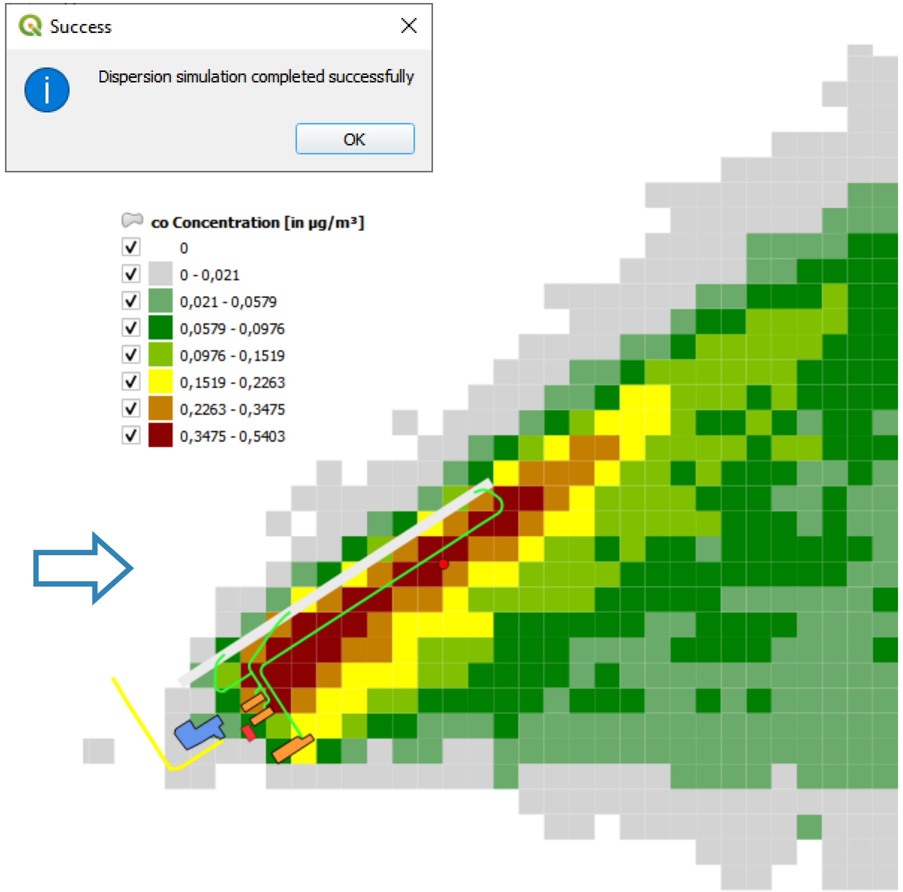
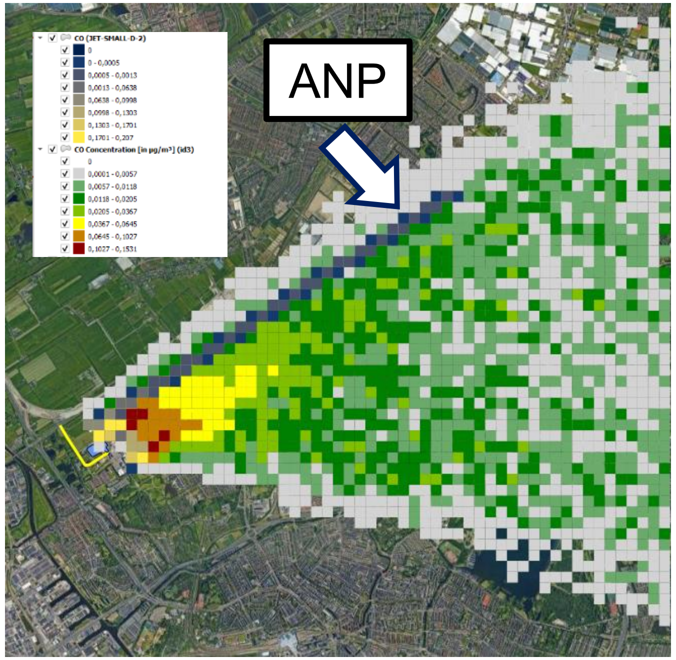


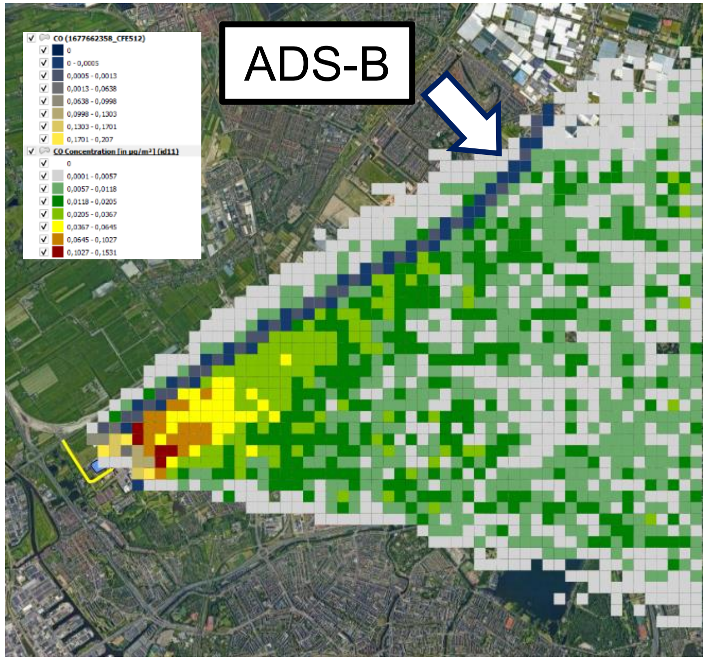
https://github.com/eurocontrol-asu/open_alaqs/blob/main/open_alaqs/assets/emissions-calculation-2.png#dispersion-modeling-with-austal

EUROCONTROL

Aircraft emissions: ANP vs ADS-B

- Possible to consider real aircraft trajectories (i.e. ADS-B data)
    - *Only in DEV version for now*
- Custom profiles must be provided by the user

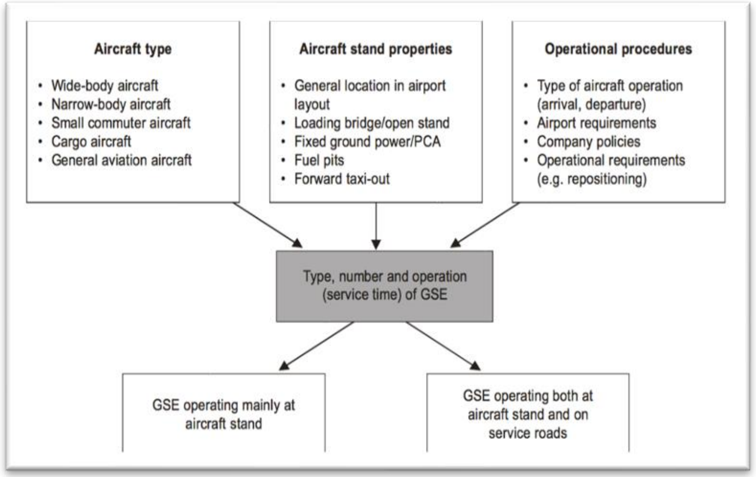




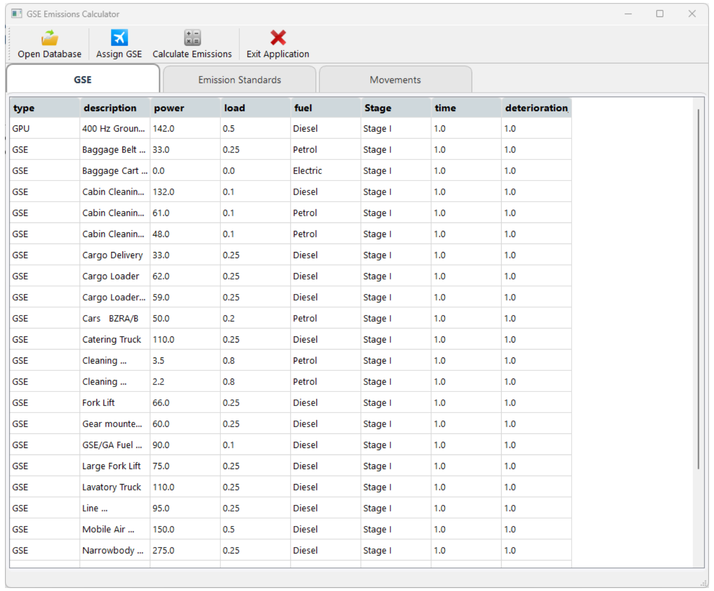
GSE custom emissions application

EUROCONTROL

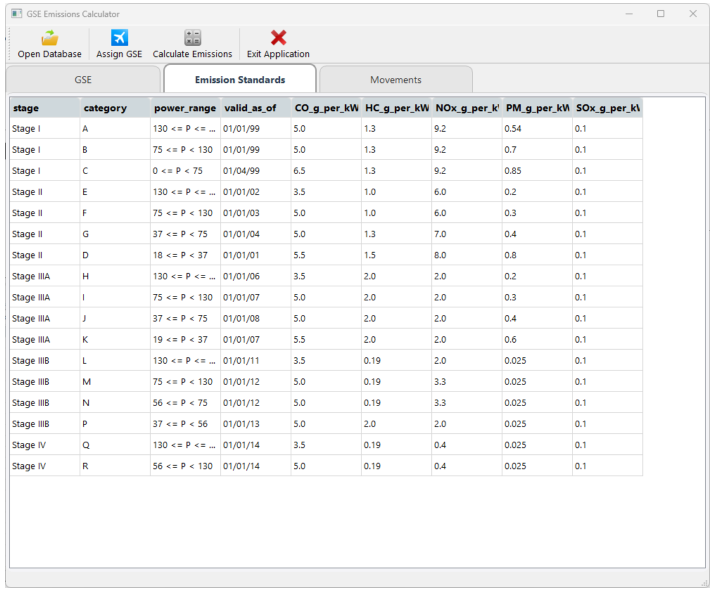
- Standalone tool to prepare custom EFs for GSE
- Default values provided – easily editable
- Methodology follows ICAO's Airport Air Quality Manual (Doc. 9889)



ICAO's Airport Air Quality Manual (Doc. 9889)



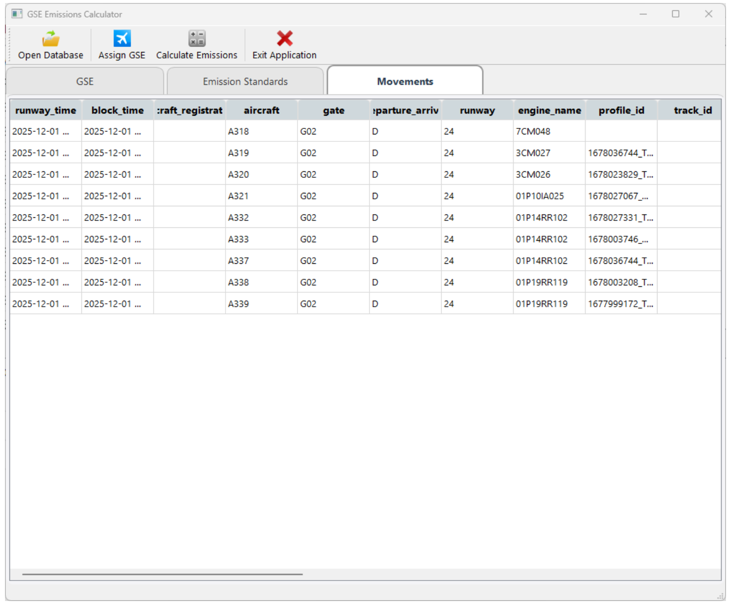
type	description	power	load	fuel	Stage	time	deterioration
GPU	400 Hz Groun...	142.0	0.5	Diesel	Stage I	1.0	1.0
GSE	Baggage Belt ...	33.0	0.25	Petrol	Stage I	1.0	1.0
GSE	Baggage Cart ...	0.0	0.0	Electric	Stage I	1.0	1.0
GSE	Cabin Cleanin...	132.0	0.1	Diesel	Stage I	1.0	1.0
GSE	Cabin Cleanin...	61.0	0.1	Petrol	Stage I	1.0	1.0
GSE	Cabin Cleanin...	48.0	0.1	Petrol	Stage I	1.0	1.0
GSE	Cargo Delivery	33.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cargo Loader	62.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cargo Loader...	59.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cars BZRA/B	50.0	0.2	Petrol	Stage I	1.0	1.0
GSE	Catering Truck	110.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cleaning ...	3.5	0.8	Petrol	Stage I	1.0	1.0
GSE	Cleaning ...	2.2	0.8	Petrol	Stage I	1.0	1.0
GSE	Fork Lift	66.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Gear mounte...	60.0	0.25	Diesel	Stage I	1.0	1.0
GSE	GSE/GA Fuel ...	90.0	0.1	Diesel	Stage I	1.0	1.0
GSE	Large Fork Lift	75.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Lavatory Truck	110.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Line ...	95.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Mobile Air ...	150.0	0.5	Diesel	Stage I	1.0	1.0
GSE	Narrowbody ...	275.0	0.25	Diesel	Stage I	1.0	1.0

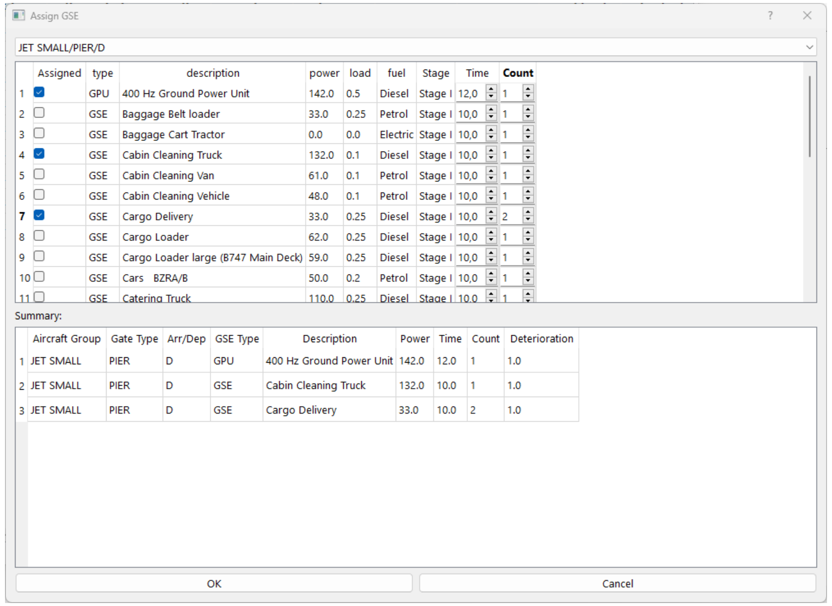


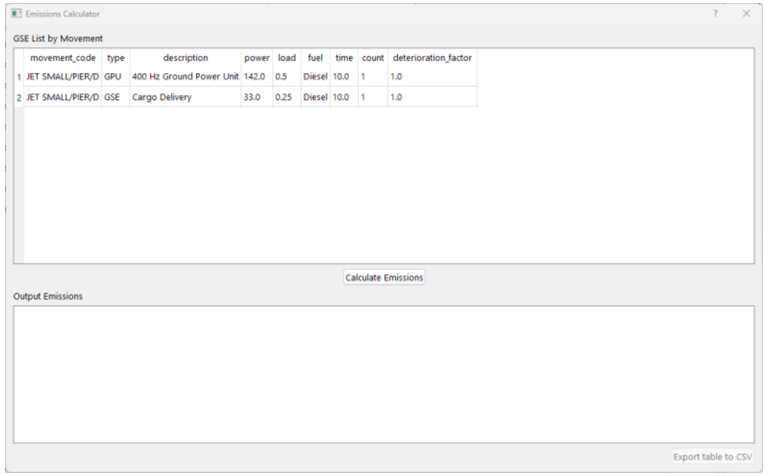
stage	category	power_range	valid_as_of	CO_g_per_kW	HC_g_per_kW	NOx_g_per_k	PM_g_per_kW	SOx_g_per_k
Stage I	A	130 <= P <= ...	01/01/99	5.0	1.3	9.2	0.54	0.1
Stage I	B	75 <= P < 130	01/01/99	5.0	1.3	9.2	0.7	0.1
Stage I	C	0 <= P < 75	01/04/99	6.5	1.3	9.2	0.85	0.1
Stage II	E	130 <= P <= ...	01/01/02	3.5	1.0	6.0	0.2	0.1
Stage II	F	75 <= P < 130	01/01/03	5.0	1.0	6.0	0.3	0.1
Stage II	G	37 <= P < 75	01/01/04	5.0	1.3	7.0	0.4	0.1
Stage II	D	18 <= P < 37	01/01/01	5.5	1.5	8.0	0.8	0.1
Stage IIIA	H	130 <= P <= ...	01/01/06	3.5	2.0	2.0	0.2	0.1
Stage IIIA	I	75 <= P < 130	01/01/07	5.0	2.0	2.0	0.3	0.1
Stage IIIA	J	37 <= P < 75	01/01/08	5.0	2.0	2.0	0.4	0.1
Stage IIIA	K	19 <= P < 37	01/01/07	5.5	2.0	2.0	0.6	0.1
Stage IIIB	L	130 <= P <= ...	01/01/11	3.5	0.19	2.0	0.025	0.1
Stage IIIB	M	75 <= P < 130	01/01/12	5.0	0.19	3.3	0.025	0.1
Stage IIIB	N	56 <= P < 75	01/01/12	5.0	0.19	3.3	0.025	0.1
Stage IIIB	P	37 <= P < 56	01/01/13	5.0	2.0	2.0	0.025	0.1
Stage IV	Q	130 <= P <= ...	01/01/14	3.5	0.19	0.4	0.025	0.1
Stage IV	R	56 <= P < 130	01/01/14	5.0	0.19	0.4	0.025	0.1

GSE custom emissions application

- Import movements from .alaqs db
- Select GSE per AC/Gate/Operation type – Number and time
- Calculate custom EFs and export/update .alaqs db







Customising OpenALAQS

Development

(Back to top)

1. Clone this repository.
2. Optionally, create a soft link between the local checkout and the QGIS plugin directory for easier developepment. (Linux instructions: `ln -s ${PWD} ${HOME}/.local/share/QGIS/QGIS3/profiles/default/python/plugins/open_alaqs/`).
3. Install `pre-commit` .
4. Develop a new feature.
5. Open a PR.
6. Wait for the CI to succeed.
7. Ensure you have a PR approval from another reviewer.
8. Merge the PR.

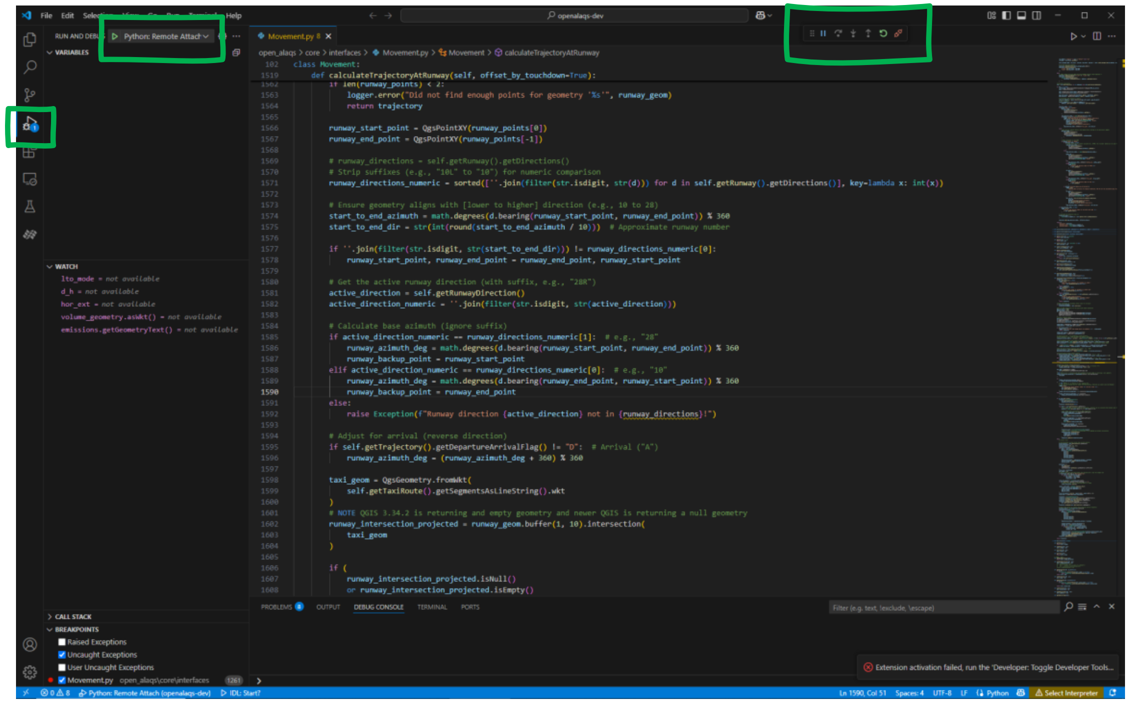
Customising OpenALAQS

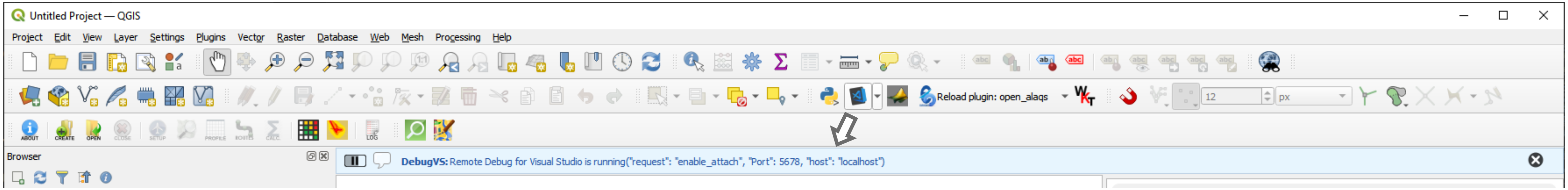
Debugging

Deugging can be done via QGIS VSCode Debug plugin and VSCode.

Sample `launch.json`:

```
{
    // Use IntelliSense to learn about possible attributes.
    // Hover to view descriptions of existing attributes.
    // For more information, visit: https://go.microsoft.com/fwlink/?linkid=830387
    "version": "0.2.0",
    "configurations": [
        {
            "name": "Python: Remote Attach",
            "type": "python",
            "request": "attach",
            "port": 5678,
            "host": "localhost",
            "pathMappings": [
                {
                    "localRoot": "${workspaceFolder}/open_alaqs",
                    "remoteRoot": "${HOME}/.local/share/QGIS/QGIS3/profiles/default/python/plugins/open_alaqs/"
                }
            ]
        }
    ]
}
```





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